

wujian100_open Userguide

Number	
Revision	1.0
Security	Public
Date	2019-10-15

Copyright © 2025 T-HEAD Semiconductor Co.,Ltd. All rights reserved.

This document is the property of T-HEAD Semiconductor Co.,Ltd. This document may only be distributed to: (i) a T-HEAD party having a legitimate business need for the information contained herein, or (ii) a non-T-HEAD party having a legitimate business need for the information contained herein. No license, expressed or implied, under any patent, copyright or trade secret right is granted or implied by the conveyance of this document. No part of this document may be reproduced, transmitted, transcribed, stored in a retrieval system, translated into any language or computer language, in any form or by any means, electronic, mechanical, magnetic, optical, chemical, manual, or otherwise without the prior written permission of T-HEAD Semiconductor Co.,Ltd.

Trademarks and Permissions

The T-HEAD Logo and all other trademarks indicated as such herein are trademarks of T-HEAD Semiconductor Co.,Ltd. All other products or service names are the property of their respective owners.

Notice

The purchased products, services and features are stipulated by the contract made between T-HEAD and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Copyright © 2025 平头哥半导体有限公司，保留所有权利。

本文档的所有权及知识产权归属于平头哥半导体有限公司及其关联公司(下称“平头哥”)。本文档仅能分派给：(i)拥有合法雇佣关系，并需要本文档的信息的平头哥员工，或(ii)非平头哥组织但拥有合法合作关系，并且其需要本文档的信息的合作方。对于本文档，未经平头哥半导体有限公司明示同意，则不能使用该文档。在未经平头哥半导体有限公司的书面许可的情形下，不得复制本文档的任何部分，传播、转录、储存在检索系统中或翻译成任何语言或计算机语言。

商标申明

平头哥的 LOGO 和其它所有商标归平头哥半导体有限公司及其关联公司所有，未经平头哥半导体有限公司的书面同意，任何法律实体不得使用平头哥的商标或者商业标识。

注意

您购买的产品、服务或特性等应受平头哥商业合同和条款的约束，本文档中描述的全部或部分产品、服务或特性可能不在您的购买或使用范围之内。除非合同另有约定，平头哥对本文档内容不做任何明示或默示的声明或保证。

由于产品版本升级或其他原因，本文档内容会不定期进行更新。除非另有约定，本文档仅作为使用指导，本文档中的所有陈述、信息和建议不构成任何明示或暗示的担保。平头哥半导体有限公司不对任何第三方使用本文档产生的损失承担任何法律责任。

平头哥半导体有限公司 T-HEAD Semiconductor Co.,LTD

地址: 杭州市西湖区西斗门路 3 号天堂软件园 A 座 15 楼 邮编: 310012

网址: www.T-head.cn



版本历史(Revisions)

[illegible]

目录(Contents)

1 System Overview	6
1.1 Functional Features	6
1.2 Address Map	7
1.2.1 Peripheral Address Map	8
1.3 PAD I/O	11
1.4 Interrupt source	14
2 Timer (TIM)	17
2.1 Register Description	17
2.1.1 Register Memory map	17
2.1.2 Register Field Description	18
2.2 Work Flow	22
3 Direct Memory Access (DMA)	23
3.1 Overview	23
3.2 Function Description	23
3.3 Register description	24
3.3.1 DMA channel base address	24
3.3.2 Address map	25
3.3.3 Register field description	26
3.4 Work Flow	34
4 Unified Serial Interface (USI)	35
4.1 Overview	35
4.2 PAD ports mapping	35
4.3 Register	35
5 Watchdog (WDT)	51
5.1 Register Description	51
5.1.1 Register Memory map	51
5.1.2 Register Field Description	52
5.2 Work Flow	56
6 Pulse Width Modulation (PWM)	57
6.1 Register Description	57
6.1.1 Register Memory Map	57
6.1.2 Register Field Description	60
6.2 Work Flow	109
7 Real-Time Clock (RTC)	111
7.1 Register Description	111
7.1.1 Register Memory Map	111
7.1.2 Registers Field Description	112
7.2 Work Flow	115
8 General-purpose I/O (GPIO)	116
8.1 Register description	116
8.1.1 Register Memory Map	116
8.1.2 Work Flow	116

2 Registers Filed Description	117
8.1.3 Work Flow	121

表格目录(Tables)

Table 1-1Memory Address Map	8
Table 1-2Peripheral Address Map	8
Table 1-3PIN List	11
Table 1-4 Interrupt source	14
Table 2-2 Timers Memory Map	17
Table 2-3 Timer1LoadCount Field Description	18
Table 2-4 Timer1CurrentValue Register Field Description	18
Table 2-5 Timer1Control Register Field Description	19
Table 2-6 Timer1 int clr Field Description	19
Table 2-7 Timer1Int Status Field Description	20
Table 2-8 Timer2LoadCount Field Description	20
Table 2-9 Timer2CurrentValue Register Field Description	21
Table 2-10 Timer2 Control Register Field Description	21
Table 2-11 Timer2 int clr Field Description	22
Table 2-12 Timer2Int Status Field Description	22
Table 3-1 DMA Memory Map for one channel	25
Table 6-1 PWMCFG Field Description	60
Table 6-2 PWMINVERTTRIG Field Description	63
Table 6-3 PWM01TRIG Field Description	64
Table 6-4 PWM23TRIG Field Description	65
Table 6-5 PWM45TRIG Field Description	65
Table 6-6 PWMNTEN1 Field Description	65
Table 6-7 PWMINTEN2 Field Description	68
Table 6-8 PWMRIS1 Field Description	71
Table 6-9 PWMRIS2 Field Description	74
Table 6-10 PWMIC1 Field Description	78
Table 6-11 PWMIC2 Field Description	80
Table 6-12 PWMIS1 Field Description	83
Table 6-13 PWMIS2 Field Description	85
Table 6-14 PWMCTL Field Description	88
Table 6-15 PWMnmLOAD Field Description	90
Table 6-16 PWMnmCOUNT Field Description	91
Table 6-17 PWMnCMP Field Description	92
Table 6-18 PWMnmDB Field Description	92
Table 6-19 CAPCTL Field Description	93
Table 6-20 CAPINTEN Field Description	95
Table 6-21 CAPRIS Field Description	97
Table 6-22 CAPIC Field Description	99
Table 6-23 CAPIS Field Description	101
Table 6-24 CAPnmT Field Description	103
Table 6-25 CAPnmMATCH Field Description	104

Tabl

e 6-26 TIMINTEN Field Description	104
Table 6-27 TIMRIS Field Description	105
Table 6-28 TIMIC Field Description	106
Table 6-29 TIMIS Field Description	107
Table 6-30 TIMnmLOAD Field Description	108
Table 6-31 TIMnmCOUNT Field Description	109
Table 6-32 CNTnmVAL Field Description	109
Table 8-1 Address Map of GPIO Registers	116
Table 8-2 gpio_output_data Field Description	117
Table 8-3 gpio_direction Field Description	117
Table 8-4 gpio_ctl Field Description	118
Table 8-5 gpio_inten Field Description	118
Table 8-6 gpio_intmask Field Description	119
Table 8-7 gpio_inttype_level Field Description	119
Table 8-8 gpio_int_polarity Field Description	120
Table 8-9 gpio_intstatus Field Description	120
Table 8-10 gpio_rawintstatus Field Description	120
Table 8-11 gpio_int_clr Field Description	121
Table 8-12 gpio_input_data Field Description	121

术语与参考资料(Abbreviations and Reference)

术语与缩略语

Abbreviations 缩略语	Full spelling 英文全名	Chinese explanation 中文解释

参考资料

编号	文 档 编号	标题	版本	日期

1 System Overview

1.1 Functional Features

- **CPU Part**
 - **32-bit general purpose CPU E902**
 - **CoreTim**
 - ◆ The circulating decrement counter is 24-bit count width
 - **VIC**
 - ◆ Supports 64 Interrupts nesting
 - ◆ Each interrupt has corresponding priority
- **Memory**
 - 64KB ISRAM and 3×64KB DSRAM on-chip SRAM
- **Characteristics of Peripherals**
 - **DMAC(x1)**
 - ◆ 16 channels
 - ◆ Supports block/group trigger mode transactions
 - **TIM (×8)**
 - ◆ connects to APB1 and APB0
 - ◆ All of timers are 32-bit count width
 - ◆ All of timers support for two operation modes: free-running and user-defined count
 - **GPIO (×1)**
 - ◆ Connects to APB
 - ◆ GPIO supports 32-bit width;
 - ◆ Each bit of GPIO supports interrupt generation
 - **USI (×3)**
 - ◆ Universal Asynchronous Receiver/Transmitter(UART)

- ◆ Inter-integrated Circuit(I2C)
- ◆ Serial Peripheral Interface(SPI)
- **RTC**
 - ◆ RTC connects to APB
 - ◆ 32-bit count width
 - ◆ Incrementing counter and comparator for interrupt generation
- **PWM**
 - ◆ Connects to APB
 - ◆ 12 input/output channels
 - ◆ 6 PWM generators, each with 1 32-bit counter, 2 PWM comparator, 1 PWM signal generator, 1 interrupt generator
 - ◆ Each PWM signal generator contains 2 channels
 - ◆ PWM output enable or disable of each PWM signal
 - ◆ Optional output inversion of each PWM signal (polarity control)
 - ◆ 6 32-bit counter and each has the following characteristics:
 - Up or Up/Down mode
 - Output frequency controlled by a 16-bit load value
 - ◆ 32-bit input capture modes
 - Input edge count mode
 - Input edge time mode
- **WDT**
 - ◆ One WDT in APB
 - ◆ Each WDT has 32 count width
 - ◆ Counter counts down from a pre-set value to zero to indicate the occurrence of a timeout

1.2 Address Map

For this chip, the memory address is divided into several parts: on-chip memory, off-chip memory, TCIP and peripherals. The detailed address map is shown as below.

Table 1-1Memory Address Map

Address Range	Size	Usage
0x0000_0000~0x0000_7FFF	64KB	Internal SRAM (inst)
0x2000_0000~0x2002_FFFF	192KB	Internal SRAM(data)
0x4000_0000~0x401F_FFFF	2MB	MAIN BUS Peripherals
0x4020_0000~0x0x7FFF_FFF F	1024M-2M	Low speed Peripherals
0x8000_0000~0x9FFF_FFFF	768M	MAIN BUS Peripherals
0xE000_E000~0xE000_EFFF	4KB	TCIP
Other	-	Reserved

1.2.1 Peripheral Address Map

The register blocks for all on-chip peripheral devices are located on 4KB boundaries. Those modules that may require additional address space are assigned additional 4KB blocks. Within a 4KB block, peripheral registers may be incompletely decoded. The description of each peripheral will define the result of accesses to unimplemented registers. For registers that do not implement all 32 bits, the unimplemented bits will return zero when read, and write to the unimplemented bits will have no effect. In general, unimplemented bits should return zero to ensure future compatibility.

Table 1-2Peripheral Address Map

Address Range	IP name	Size	Master/ Slave	Description
	E902		M0/M2	Core
	DMA		M3	DMA
	MDummy0		M4	main_mdumy_top0
	MDummy1		M5	main_mdumy_top1
	MDummy2		M6	main_mdumy_top2
0x0000_0000~0x0000_27F F	ISRAM	64K	S0	ROM
0x1000_0000~0x1007_FFF F	MemDummy	512K	S1	instmem_dummy top0
0x2000_0000~0x2000_FFF F	SRAM	64K	S2	DATA SRAM
0x2001_0000~0x2001_FFF F	SRAM	64K	S3	DATA SRAM

0x2002_0000~0x2002_FFF F	SRAM	64K	S4	DATA SRAM
0x3000_0000~0x3007_FFF F	MemDummy	512K	S5	datamem_dummy_top1
0x4000_0000~0x4000_3FF F	DMA	16K	S6	DMA Controller
0x4001_0000~0x4001_FFF F	Dummy	64K	S7	main_dummy_top0
0x4002_0000~0x4002_FFF F	Dummy	64K	S8	main_dummy_top1
0x4010_0000~0x401F_FFF F	Dummy	1M	S9	main_dummy_top2
0x4020_0000~0x7FFF_FFF F	LSBUS	1024M- 2M	S10	AHB LS BUS
0x8000_0000~0x9FFF_FFF F	Dummy	512M	S11	main_dummy_top3
Other	REV	-	REV	Reserved
AHB LS BUS				
0x4020_0000~0x4020_0FF F	Dummy	4K	S0	lsbus_dummy_top0
0x4030_0000~0x403F_FFF F	Dummy	1M	S1	lsbus_dummy_top1
0x5000_0000~0x5004_FFF F	APB0	320K	S2	APB0
0x6000_0000~0x6004_FFF F	APB1	320K	S3	APB1
0x7000_0000~0x77FF_FFF F	Dummy	128M	S4	lsbus_dummy_top2
0x7800_0000~0x7FFF_FFF F	Dummy	128M	S5	lsbus_dummy_top3
APB0				
0x5000_0000~0x5000_03F F	TIM0	1K	P0	Timer 0
0x5000_0400~0x5000_07F F	TIM2	1K	P1	Timer 2
0x5000_0800~0x5000_0BF F	TIM4	1K	P2	Timer 4
0x5000_0C00~0x5000_0FF F	TIM6	1K	P3	Timer 6
0x5002_8000~0x5002_8FF F	USI0	16K	P4	USI0

0x5002_9000~0x5002_9FF F	USI2	16K	P5	USI2
0x5000_4000~0x5000_7FF F	Dummy	16K	P6	apb0_dummy_top1
0x5000_8000~0x5000_BFF F	WDT	16K	P7	WDT
0x5000_C000~0x5000_FFF F	Dummy	16K	P8	apb0_dummy_top2
0x5001_0000~0x5001_3FF F	Dummy	16K	P9	apb0_dummy_top3
0x5001_4000~0x5001_7FF F	Dummy	16K	P10	apb0_dummy_top4
0x5001_8000~0x5001_BFF F	Dummy	16K	P11	apb0_dummy_top5
0x5001_C000~0x5001_FFF F	PWM	16K	P12	PWM
0x5002_0000~0x5002_3FF F	Dummy	16K	P13	apb0_dummy_top7
0x5002_4000~0x5002_7FF F	Dummy	16K	P14	apb0_dummy_top8
0x5003_0000~0x5003_3FF F	Dummy	16K	P15	apb0_dummy_top9
APB1				
0x6000_0000~0x6000_03F F	TIM1	1K	P0	Timer 1
0x6000_0400~0x6000_07F F	TIM3	1K	P1	Timer 3
0x6000_0800~0x6000_0BF F	TIM5	1K	P2	Timer 5
0x6000_0C00~0x6000_0FF F	TIM7	1K	P3	Timer 7
0x6002_8000~0x6002_BFF F	USI1	16K	P4	USI1
0x6001_8000~0x6001_BFF F	GPIO	16K	P5	GPIO
0x6000_4000~0x6000_7FF F	RTC	16K	P6	RTC
0x6000_8000~0x6000_BFF F	Dummy	16K	P7	apb1_dummy_top1



0x6000_C000~0x6000_FFF F	Dummy	16K	P8	apb1_dummy_top2
0x6001_0000~0x6001_3FF F	Dummy	16K	P9	apb1_dummy_top3
0x6001_4000~0x6001_7FF F	Dummy	16K	P10	apb1_dummy_top4
0x6001_C000~0x6001_FFF F	Dummy	16K	P11	apb1_dummy_top5
0x6002_0000~0x6002_3FF F	Dummy	16K	P12	apb1_dummy_top6
0x6002_4000~0x6002_7FF F	Dummy	16K	P13	apb1_dummy_top7
0x6002_C000~0x6002_FFF F	Dummy	16K	P14	apb1_dummy_top8
0x6003_0000~0x6003_3FF F	PMU	16K	P15	Power Management Unit(dummy)

1.3 PAD I/O

Each of IO can be shared by less than 4 peripherals at one time. The details of AFIO function as follows.

Table 1-3PIN List

Num	Pin name	I/O	Width	Description
1	PIN_EHS	I	1	External high speed osc clock
2	POUT_EHS	O	1	
3	PIN_ELS	I	1	External low speed osc clock
4	POUT_ELS	O	1	
5	PAD_MCURST	I	1	PAD system reset , active low
6	PAD_JTAG_TCLK	I	1	CPU JTAG TCLK
7	PAD_JTAG_TMS	I/O	1	CPU JTAG TMS
8	PAD_GPIO_0	I/O	1	GPIO0_0 signal

9	PAD_GPIO_1	I/O	1	GPIO0_1 signal
10	PAD_GPIO_2	I/O	1	GPIO0_2 signal
11	PAD_GPIO_3	I/O	1	GPIO0_3 signal
12	PAD_GPIO_4	I/O	1	GPIO0_4 signal
13	PAD_GPIO_5	I/O	1	GPIO0_5 signal
14	PAD_GPIO_6	I/O	1	GPIO0_6 signal
15	PAD_GPIO_7	I/O	1	GPIO0_7 signal
16	PAD_GPIO_8	I/O	1	GPIO0_8 signal
17	PAD_GPIO_9	I/O	1	GPIO0_9 signal
18	PAD_GPIO_10	I/O	1	GPIO0_10 signal
19	PAD_GPIO_11	I/O	1	GPIO0_11 signal
20	PAD_GPIO_12	I/O	1	GPIO0_12 signal
21	PAD_GPIO_13	I/O	1	GPIO0_13 signal
22	PAD_GPIO_14	I/O	1	GPIO0_14 signal
23	PAD_GPIO_15	I/O	1	GPIO0_15 signal
24	PAD_GPIO_16	I/O	1	GPIO0_16 signal
25	PAD_GPIO_17	I/O	1	GPIO0_17 signal
26	PAD_GPIO_18	I/O	1	GPIO0_18 signal
27	PAD_GPIO_19	I/O	1	GPIO0_19 signal
28	PAD_GPIO_20	I/O	1	GPIO0_20 signal
29	PAD_GPIO_21	I/O	1	GPIO0_21 signal



30	PAD_GPIO_22	I/O	1	GPIO0_22 signal
31	PAD_GPIO_23	I/O	1	GPIO0_23 signal
32	PAD_GPIO_24	I/O	1	GPIO0_24 signal
33	PAD_GPIO_25	I/O	1	GPIO0_25 signal
34	PAD_GPIO_26	I/O	1	GPIO0_26 signal
35	PAD_GPIO_27	I/O	1	GPIO0_27 signal
36	PAD_GPIO_28	I/O	1	GPIO0_28 signal
37	PAD_GPIO_29	I/O	1	GPIO0_29 signal
38	PAD_GPIO_30	I/O	1	GPIO0_30 signal
39	PAD_GPIO_31	I/O	1	GPIO0_31 signal
40	PAD_PWM_CH0	I/O	1	PWM channel0 signal
41	PAD_PWM_CH1	I/O	1	PWM channel1 signal
42	PAD_PWM_CH2	I/O	1	PWM channel2 signal
43	PAD_PWM_CH3	I/O	1	PWM channel3 signal
44	PAD_PWM_CH4	I/O	1	PWM channel4 signal
45	PAD_PWM_CH5	I/O	1	PWM channel5 signal
46	PAD_PWM_CH6	I/O	1	PWM channel6 signal
47	PAD_PWM_CH7	I/O	1	PWM channel7 signal
48	PAD_PWM_CH8	I/O	1	PWM channel8 signal
49	PAD_PWM_CH9	I/O	1	PWM channel9 signal
50	PAD_PWM_CH10	I/O	1	PWM channel10 signal

51	PAD_PWM_CH11	I/O	1	PWM channel11 signal
52	PAD_USI0_NSS	I/O	1	USI0 Slave select
53	PAD_USI0_SCLK	I/O	1	USI0 Serial clock
54	PAD_USI0_SD0	I/O	1	USI0 Serial data0
55	PAD_USI0_SD1	I/O	1	USI0 Serial data1
56	PAD_USI1_NSS	I/O	1	USI1 Slave select
57	PAD_USI1_SCLK	I/O	1	USI1 Serial clock
58	PAD_USI1_SD0	I/O	1	USI1 Serial data0
59	PAD_USI1_SD1	I/O	1	USI1 Serial data1
60	PAD_USI2_NSS	I/O	1	USI2 Slave select
61	PAD_USI2_SCLK	I/O	1	USI2 Serial clock
62	PAD_USI2_SD0	I/O	1	US2 Serial data0
63	PAD_USI2_SD1	I/O	1	US2 Serial data1

1.4 Interrupt source

Table 1-4 Interrupt source

Number	Interrupt Source
6:0	REV
1	CoreTim
15:8	REV
16	GPIO0
18:17	TIM0[1:0]
20:19	TIM1[1:0]

22:21	TIM2[1:0]
24:23	TIM3[1:0]
25	PWM
26	RTC
27	WDT
28	USI0
29	USI1
30	USI2
31	PMU
32	DMAC0
34:33	TIM4[1:0]
36:35	TIM5[1:0]
38:37	TIM6[1:0]
40:39	TIM7[1:0]
41	IMEMDUMMY0 DMEMDUMMY0
42	MAIN_DUMMY0
43	MAIN_DUMMY1
44	MAIN_DUMMY2
45	MAIN_DUMMY3
46	LSBUS_DUMMY0
47	LSBUS_DUMMY1
48	LSBUS_DUMMY2
49	LSBUS_DUMMY3

50	APB0_DUMMY1
51	APB0_DUMMY2
52	APB0_DUMMY3
53	APB0_DUMMY4
54	APB0_DUMMY5
55	APB0_DUMMY7
56	APB0_DUMMY8
57	APB0_DUMMY9
58	APB1_DUMMY1
59	APB1_DUMMY2
60	APB1_DUMMY3
61	APB1_DUMMY4
62	APB1_DUMMY5 APB1_DUMMY6
63	APB1_DUMMY7 APB1_DUMMY8

2 Timer (TIM)

2.1 Register Description

The following tables describe the memory map of the register in the Timers.

2.1.1 Register Memory map

In each TIMER IP , two timers are provided:Timer1 and Timer2.Following table describes the registers address map of Timer1 and the Timer2.

Table 2-2 Timers Memory Map

Name	Address offset	Width	Access	Reset Value	Description
Timer1LoadCount	0x00	32	R/W	32'b0	Value to be loaded into Timer1.
Timer1CurrentValue	0x04	32	R	32'b0	Current Value of Timer1.
Timer1Control Reg	0x08	4	R/W	4'b0	Control Register for Timer1.
Timer1_int_clr	0x0C	1	R	1'b0	Clears the interrupt from Timer1.
Timer1Int Status	0x10	1	R	1'b0	Contains the interrupt status for Timer1.
Timer2LoadCount	0x14	32	R/W	32'b0	Value to be loaded into Timer2.
Timer2CurrentValue	0x18	32	R	32'b0	Current Value of Timer2



Timer2Control Reg	0x1c	4	R/W	4'b0	Control Register for Timer2
Timer2_int_clr	0x20	1	R	1'b0	Clears the interrupt from Timer2
Timer2Int Status	0x24	1	R	1'b0	Contains the interrupt status for Timer2

2.1.2 Register Field Description

The following section mainly describes the Timer1 register field and the Timers system register field. Timer2 register field description just replicates that of Timer1 using the address offset specified in above table.

Timer1 Load Count

Address Offset: 0x00

Read/write access: read/write

Table 2-3 Timer1LoadCount Field Description

Bits	Name	R/W	Description
31:0	Timer1 Load Count Register	R/W	Value to be loaded into Timer1. This is the value from which counting commences. Any value written to this register is loaded into the associated timer. Reset Value: 32'b0

Timer1 Current Value

Address Offset: 0x04

Read/write access: read

Table 2-4 Timer1CurrentValue Register Field Description

Bits	Name	R/W	Description
31:0	Timer1 Current Value Register	R	Current Value of Timer1. Reset Value: 32'b0

Timer1 Control Reg

Address Offset: 0x08

Read/write access: read/write

Table 2-5 Timer1Control Register Field Description

Bits	Name	R/W	Description
31:5	Reserved, read as zero		
4	Timer Hardware Trigger enable	R/W	0 (timer hardware trigger disable) 1 (timer hardware trigger enable)
3	Reserved, read as zero		
2	Timer Interrupt Mask	R/W	0 (timer interrupt not masked) 1 (timer interrupt masked) Reset Value: 1'b0
1	Timer Mode Select	R/W	0 (free-running) 1 (user-defined running) Reset Value: 1'b0
0	Timer Enable Select	R/W	0 (disabled) 1 (enabled) Reset Value: 1'b0

Timer1_int_clr

Address Offset: 0x0c

Read/write access: read

Table 2-6 Timer1 int clr Field Description

Bits	Name	R/W	Description
31:1	Reserved, read as zero		

0	Timer1 int clr Register	R	Reading from this register returns all zeros(0) and clears the interrupt from Timer1
---	----------------------------	---	---

Timer1Int Status

Address Offset: 0x10

Read/write access: read

Table 2-7 Timer1Int Status Field Description

Bits	Name	R/W	Description
31:1	Reserved, read as zero		
0	Timer1Int Status Register	R	Contains the interrupt status for Timer1

Timer2 Load Count

Address Offset: 0x14

Read/write access: read/write

Table 2-8 Timer2LoadCount Field Description

Bits	Name	R/W	Description
31:0	Timer2 Load Count Register	R/W	Value to be loaded into Timer2. This is the value from which counting commences. Any value written to this register is loaded into the associated timer. Reset Value: 32'b0

Timer2 Current Value

Address Offset: 0x18

Read/write access: read

Table 2-9 Timer2CurrentValue Register Field Description

Bits	Name	R/W	Description
31:0	Timer2 Current Value Register	R	Current Value of Timer2 Reset Value: 32'b0

Timer2 Control Reg

Address Offset: 0x1c

Read/write access: read/write

Table 2-10 Timer2 Control Register Field Description

Bits	Name	R/W	Description
31:5	Reserved, read as zero		
4	Timer Hardware Trigger enable	R/W	0 (timer hardware trigger disable) 1 (timer hardware trigger enable)
3	Reserved, read as zero		
2	Timer Interrupt Mask	R/W	0 (timer interrupt not masked) 1 (timer interrupt masked) Reset Value: 1'b0
1	Timer Mode Select	R/W	0 (free-running) 1 (user-defined running) Reset Value: 1'b0
0	Timer Enable Select	R/W	0 (disabled) 1 (enabled) Reset Value: 1'b0

Timer2 int clr

Address Offset: 0x20

Read/write access: read

Table 2-11 Timer2 int clr Field Description

Bits	Name	R/W	Description
31:1	Reserved, read as zero		
0	Timer2 int clr Register	R	Reading from this register returns all zeros(0) and clears the interrupt from Timer2

Timer2Int Status

Address Offset: 0x24

Read/write access: read

Table 2-12 Timer2Int Status Field Description

Bits	Name	R/W	Description
31:1	Reserved, read as zero		
0	Timer2 Interrupt Status Register	R	Contains the interrupt status for Timer2. Reset Value: 32'b0

2.2 Work Flow

Normal program of the timer is as follows:

1. Disable the timer and program its operating mode by writing to its control register, TimerN Control Reg.
2. Load the timer's Timer Load Count register, TimerN Load Count.
3. Enable the timer by writing the TimerN Control Reg register.

3 Direct Memory Access (DMA)

3.1 Overview

DMAC is compliant with AMBA2.0 AHB-lite Specification, Support max 16 channels. Every channel only supports block trigger mode.

There are mask registers to mask the interrupts and clean registers to clear the interrupts. Block transfer completion interrupt will generate after all block transfer done.

3.2 Function Description

- 16 channels, each channel has a source addr and a dest addr, data will be transferred from source addr to dest addr, both the source addr and the dest addr can be programmed
- Support 16 priority level, Channel 0 has highest priority than other channels.
- Support enable protect, DMA config can not be changed once enable is set to 1.
- The source and dest Address have 3 types: increase, decrease, or no change
- Programmable block transaction size for each channel with max 4096bytes
- Support source and destination transaction with different width, 8/16/32bits can be selected
- Programmable enable and disable of DMA channel
- The low priority channel can be interrupt by high priority channel

- Data little-endian or big-endian
- Interrupt , Interrupt can be generated when:

a.Block transfer completion

b.Half Block transfer completion

c.Trigger event finish

d.Error condition

Each interrupt can be masked or cleared by registers.

3.3 Register description

3.3.1 DMA channel base address

Include 17 base address: (16 Channel registers and 1 global registers)

Channel 0 registers include registers(internal base offset address is:0x0) : SAR0 , DAR0 , CH0_CTRL_A , CH0_CTRL_B , CH0_INT_MASK , CH0_INT_STATUS , CH0_INT_CLEAR , CH0_SOFT_REQ , CH0_EN

Channel 1 registers include registers(internal base offset address is:0x30) : SAR1 , DAR1 , CH1_CTRL_A , CH1_CTRL_B , CH1_INT_MASK , CH1_INT_STATUS , CH1_INT_CLEAR , CH1_SOFT_REQ , CH1_EN

.....

Channel n registers include registers(internal base offset address is:0x30*n) : SARn , DARn , CHn_CTRL_A , CHn_CTRL_B , CHn_INT_MASK , CHn_INT_STATUS , CHn_INT_CLEAR , CHn_SOFT_REQ , CHn_EN

.....

Channel 15 registers include registers(internal base offset address is:0x2D0) : SAR15 , DAR15 , CH15_CTRL_A , CH15_CTRL_B , CH15_INT_MASK , CH15_INT_STATUS , CH15_INT_CLEAR , CH15_SOFT_REQ , CH15_EN

16 channels internal offset address is the same.

Global register (internal base offset address is :0x330):CHSR,DMACCFG

Channel 0 base address	0x0	Channel 1 base address	0x30
Channel 2 base address	0x60	Channel 3 base address	0x90
Channel 4 base address	0xC0	Channel 5 base address	0xF0



Channel 6 base address	0x120	Channel 7 base address	0x150
Channel 8 base address	0x180	Channel 9 base address	0x1B0
Channel 10 base address	0x1E0	Channel 11 base address	0x210
Channel 12 base address	0x240	Channel 13 base address	0x270
Channel 14 base address	0x2A0	Channel 15 base address	0x2D0
Global register base address	0x330	-	-

3.3.2 Address map

The following table describe the register memory map for one channel

Table 3-1 DMA Memory Map for one channel

Register	Offset	Width	Describe	Reset Value
SAR _n	0x0	32 bits	Channel n Source Address Register	0x0000_0000
DAR _n	0x4	32 bits	Channel n Destination Address Register	0x0000_0000
CH _n _CTRL_A	0x8	32 bits	Channel n Control Register A	0x0000_0000
CH _n _CTRL_B	0xc	32 bits	Channel n Control Register B	0x0000_0000
CH _n _INT_MASK	0x10	32 bits	Channel n Mask Interrupt Register	0x0000_0000
CH _n _INT_STATUS	0x14	32 bits	Channel n Status Interrupt Register	0x0000_0000
CH _n _INT_CLEAR	0x18	32 bits	Channel n Clear Interrupt Register	0x0000_0000



CHn_SOFT_REQ	0x1c	32 bits	Channel n Software Handshaking Request Register	0x0000_0000
CHn_EN	0x20	32 bits	Channel n Enable Control Register	0x0000_0000

Every channel has a base of offset address which is described in 3.3.1.

DMAC global registers base address is 0x330,these registers are the configs or status of all 16 channels. Such as enable, busy status or pending state

Register	Offset	Width	Describe	Reset Value
Reserved	0x0	-	-	-
Reserved	0x4	-	-	-
CHSR	0x8	32 bits	Channel busy status register	0x0000_0000
DMACCFG	0xC	32bits	DMAC configure register	0x0000_0000

3.3.3 Register field description

3.3.3.1 Channel n Source Address Register (SARn)

Address Offset: 0x0

bits	R/W	Describe	
31:0	R/W	SARN[31:0]	Channel n Source read Address Register reset value:0x0

3.3.3.2 Channel n Destination Address Register (DARN)

Address Offset: 0x4

bits	R/W	Describe	
31:0	R/W	DARN[31:0]	Channel n Destination write Address Register reset value:0x0

3.3.3.3 Channel n Control Register A (CHn_CTRL_A)

Address Offset: 0x8

bits	R/W	Describe	
31:24	R/W	-	Reserved
23:12	R/W	BLOCK_TL[11:0]	Block Transfer Length writes this field before the channel is enabled in order to indicate the block Length (It's bytes total number : BLOCK_TL+1 it's max length is 4096 bytes) reset value:0x0
11 : 8	R/W	-	Reserved
7:6	R/W	SINC[1:0]	Source Address Increment Indicates whether to increment or decrease the source address on every transfer. 00:increment 01: decrease 1x: no change reset value: 0x0
5:4	R/W	DINC[1:0]	Destination Address Increment Indicates whether to increment or decrease the destination address on every transfer. 00: increment 01: decrease 1x: no change reset value: 0x0



3:2	R/W	SRC_TR_WIDTH[1:0]	Source Transfer Width mapped to AHB bus “hsize.” 00: 8bit 01: 16bit 10: 32bit 11 : reserved reset value: 0x0
1:0	R/W	DST_TR_WIDTH[1:0]	Destination Transfer Width mapped to AHB bus “hsize.” 00: 8bit 01: 16bit 10: 32bit 11 : reserved reset value: 0x0

3.3.3.4 Channel n Control Register B (CHn_CTRL_B)

Address Offset: 0xc

bits	R/W	Describe	
31:19		Reserved	
18:15	R/W	PROTCTL[3:0]	Channel n Protection Control used PROTCTL[3:0] to drive the AHB HPROT. PROTCTL[2]:controls the secure access. 1:secure access(It only support secure access) 0:normal access(It support secure/no secure access) reset value:0x0

14	R/W	DSTDTLGC	Destination Write data Little-Big Endian change control 0: little endian 1: big endian reset value:0x0 Detail information can be found Function Description
13	R/W	SRCDTLGC	Source read data Little-Big Endian change control 0: little endian 1: big endian reset value:0x0 Detail information can be found Function Description
12:3	Reserved		
2:1	R/W	TRGTMDC[1:0]	Trigger transfer mode control register 2'b00:rev 2'b01:rev 2'b1x:Block trigger mode Reset value:0x0 These bits should be set to 2'b10 or 2'b11
0	R/W	INT_EN	Interrupt Enable 0: all interrupt-generating sources are disabled. 1: all interrupt-generating sources are enabled. reset value: 0x0

3.3.3.5 Channel n Mask Interrupt Register (CHn_INT_MASK)

Address Offset:0x10

bits	R/W	Describe
------	-----	----------

3	R/W	masktrgetcmpfr	Mask for trigger event complete Interrupt of channel n 0: mask 1: not mask reset value: 0x0
2	R/W	maskhtfr	Mask for transfer half complete Interrupt of channel n 0: mask 1: not mask reset value: 0x0
1	R/W	masktfr	Mask for transfer complete Interrupt of channel n 0: mask 1: not mask reset value: 0x0
0	R/W	maskErr	Mask for Transfer Error Interrupt of Channel n 0: mask 1: not mask reset value: 0x0

3.3.3.6 Channel n Status Interrupt Register (CHn_INT_STATUS)

Address Offset:0x14

bits	R/W	Describe	
3	R	statustrgetcmpfr	Status for Transfer Trigger event complete . Reset value:0x0
2	R	statushtfr	Status for Transfer half of block length Complete Interrupt of Channel n reset value: 0x0
1	R	statustfr	Status for Transfer Block Length Complete Interrupt of Channel n reset value: 0x0

0	R	statusErr	Status for Transfer Error Interrupt of Channel n(It's only detect AHB bus HRESP error condition) reset value: 0x0
---	---	-----------	--

3.3.3.7 Channel n Clear Interrupt Register (CHn_INT_CLEAR)

Address Offset:0x18

bits	R/W	Describe	
3	W	cleartrgetcmpfr	Clear for statustrgetcmpif 0: no clear 1: clear reset value: 0x0
2	W	clearhtfr	Clear for statushtfr 0: no clear 1: clear reset value: 0x0
1	W	cleartfr	Clear for Statustfr 0: no clear 1: clear reset value: 0x0
0	W	clearErr	Clear for StatusErr 0: no clear 1: clear reset value: 0x0

3.3.3.8 Channel n Software Handshaking Request Register (CHn_SOFT_REQ)

Address Offset:0x1c

bits	R/W	Describe
------	-----	----------

0	W	soft_req	Software Request Register reset value: 0x0
---	---	----------	---

3.3.3.9 Channel n Enable control register (CHn_EN)

Address Offset:0x20

bits	R/W	Describe	
0	R/W	chn_en	Channel 0 Enable Register When this bit set 1,the ADRn , SARn,CHANn_CTRLA,CHANn_CTRLB can't be modified. reset value: 0x0 After transfer is over,this bit will auto cleared to 0 by hardware

3.3.3.10 Global registers

Channel busy status register

bits	R/W	Describe	
23:16	-	-	Reserved
15	R	Ch15bsy	Channel 15 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
14	R	Ch14bsy	Channel 14 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
13	R	ch12bsy	Channel 13 busy ,It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid

12	R	Ch12bsy	Channel 12 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
11	R	Ch11bsy	Channel 11 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
10	R	ch10bsy	Channel 10 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
9	R	Ch9bsy	Channel 9 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
8	R	Ch8bsy	Channel 8 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
7	R	Ch7bsy	Channel 7 busy ,It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
6	R	Ch6bsy	Channel 6 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
5	R	Ch5bsy	Channel 5 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
4	R	Ch4bsy	Channel 4 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid

3	R	Ch3bsy	Channel 3 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
2	R	ch2bsy	Channel 2 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
1	R	ch1bsy	Channel 1 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid
0	R	ch0bsy	Channel 0 busy , It owns the data bus 1 : data transfer busy valid 0 : data transfer busy invalid

DMACCFG

bits	R/W	Describe	
31:1	-	-	Reserved
0	R/W	DMACEN	Global DMAC enable. After this bit is set ,all channels can be work.

3.4 Work Flow

1.Configure register: SARn,DARn,CHn_CTRL_A,CHn_CTRL_B ,and any other register.And set DMACCFG.DMACEN and CHn_EN.chnen 1'b1 at last.

2.Configure soft_req for trigger transfer.

3.Wait int flag,The CHn_INT_STATUS.statustrf interrupt flag will clear CHn_EN.chn_en to 0.

4.Afer receive CHn_INT_CLEAR.cleartrf clear interrupt flag.

Note:SARn,DARn,CHn_CTRL_A,CHn_CTRL_B is enable protected(use channel internal enable protect ,not DMACEN) .Before set DMACCFG.DMACEN 1'b1,user should configure it's channel registers.

4

Unified Serial Interface (USI)

4.1 Overview

The unified serial interface (USI) module is a serial interface useful for communication with other peripheral or microcontroller devices by the serial interface in USI. There are APB, DMA interface and interrupt interface in USI, it can be integrated in SOC as an APB device easily, transfer data with DMA and interrupt to CPU by the interrupt interface. The USI module can be configured as one function of UART or I²C or SPI. The USI's function and the value of control registers in USI cannot be changed when the USI is working, Otherwise the data may be lost. The change can take place when the USI module is disable.

4.2 PAD ports mapping

PAD ports mapping with internal serial module's ports:

Serial Ports(PAD)	UART	I2C	SPI
SCLK	RXD	SCL	SCK
SD0	TXD	SDA	MOSI
SD1	CTS		MISO
NSS	RTS		NSS

4.3 Register

Register and field descriptions

USI_CTRL

- Name: USI module enable control register.
- Address offset: 0x000

All of sub modules (TX FIFO, RX FIFO, UART, I²C and SPI) are controlled by this register.

Field	Name	Default	Access	Description
31: 4		0x0	RO	Reserved.
3	RX_FIFO_EN	0x0	R/W	RX FIFO enable controller. 0x0: disable RX FIFO. 0x1: enable RX FIFO.



2	TX_FIFO_EN	0x0	R/W	TX FIFO enable controller. 0x0: disable TX FIFO. 0x1: enable TX FIFO.
1	FM_EN	0x0	R/W	Function modules (UART I2C SPI) enable controller. 0x0: disable function modules. 0x1: enable function modules.
0	USI_EN	0x0	R/W	All modules in USI enable controller. 0x0: disable USI module. 0x1: enable USI module.

MODE_SEL

- Name: mode select control register.
- Address offset: 0x004

This register is used to select one module of three sub modules (UART I2C and SPI).

Field	Name	Default	Access	Description
31: 2		0x0	RO	Reserved.
1: 0	MODE_SEL	0x0	R/W	0x0: UART. 0x1: I ² C. 0x2: SPI. 0x3: mode keep the last set value unchanged.

TX_FIFO

- Name: the transmitter FIFO
- Address offset: 0x008

The TX data is stored in this FIFO, then the USI will read from this FIFO, and send the data.

Field	Name	Default	Access	Description
31: 16		0x0	RO	Reserved.
15: 0	TX_DATA		WO	Transmitting data

RX_FIFO

- Name: the receiver FIFO
- Address offset: 0x008

If USI received the data, the USI will store the data into RX FIFO.

Field	Name	Default	Access	Description
31: 16		0x0	RO	Reserved.
15: 0	RX_DATA		RO	Received data

FIFO_STA

- Name: the status of receive FIFO
- Address offset: 0x00C

This register lists the FULL or EMPTY data counter status for TX and RX FIFO.

Field	Name	Default	Access	Description
31: 20		0x0	RO	Reserved.
19: 16	RX_NUM	0x0	RO	The counter indicates how many data word is stored in the RX FIFO.
15: 12		0x0	RO	Reserved.
11: 8	TX_NUM	0x0	RO	The counter indicates how many data words is stored in the TX FIFO.
7: 4		0x0	RO	Reserved.
3	RX_FULL	0x0	RO	0x0: the TX FIFO is not full. 0x1: the RX FIFO is full.
2	RX_EMPTY	0x1	RO	0x0: the RX FIFO is not empty. 0x1: the RX FIFO is empty
1	TX_FULL	0x0	RO	0x0: the TX FIFO is not full 0x1: TX FIFO is full
0	TX_EMPTY	0x1	RO	0x0: the TX FIFO is not empty. 0x1: the TX FIFO is empty

CLK_DIV0

- Name: I2C SCL high counter register.
- Address offset: 0x010

Description: The clock divider register. In UART mode, the register set the baud rate; in I²C mode, the register sets the SCL clock high-period count; in the SPI mode, the register sets the SCK clock period count.

Field	Name	Default	Access	Description
31: 24		0x0	RO	Reserved.
23: 0	CLK_DIV0	0x20	R/W	System clock divided register. UART: baud rate. Baud rate = $F_{PCLK}/(16 * (CLK_DIV0+1))$ I2C: the register sets SCL clock high-period count. $F_{SCL} = F_{PCLK}/(CLK_DIV0+CLK_DIV1+2)$ SPI: divider for SCK clock. $F_{SCK} = F_{PCLK}/CLK_DIV0$, CLK_DIV0 is even value

CLK_DIV1

- Name: I2C SCL low counter register.
- Address offset: 0x014

Description: This register is only for I²C, the register set the I²C SCL clock low-period count.



Field	Name	Default	Access	Description
31: 24		0x0	RO	Reserved.
23: 0	CLK_DIV1	0x30	R/W	I2C: sets SCL clock low-period counter. UART, SPI: no used.

UART_CTRL

- Name: UART control register
- Address offset: 0x018

Description: All of the features (Data length, STOP bit length and parity) can be configured by this register.

Field	Name	Default	Access	Description
31: 6		0x0	RO	Reserved.
5	EPS	0x0	R/W	Even parity select. This bit is used to select between even and odd parity, when parity is enable (PEN set to 1). 0x0: an odd number of logic 1s is transmitted or checked. 0x1: an even number of logic 1s is transmitted or checked.
4	PEN	0x0	R/W	Parity enable. This bit is used to enable and disable parity generation and detection in transmitted and received serial character respectively. 0x0: parity disable. 0x1: parity enable.
3: 2	PBIT	0x0	R/W	Number of stop bits. This is used to select the number of stop bits per character that the peripheral transmits and receives. 0x0: 1 stop bit. 0x1: 1.5 stop pbits. 0x2: 2 stop bits. 0x3: number of stop bits keep the last set value unchanged.



1: 0	DBIT	0x3	R/W	Data bits select. This is used to select the number of data bits per character that the peripheral transmits and receive. The number of bit that may be selected areas follows: 0x0: 5-bit. 0x1: 6-bit. 0x2: 7-bit. 0x3: 8-bit.
------	------	-----	-----	---

UART_STA

- Name: UART status register.
- Address offset: 0x01C

Description: The RXD and TXD working state can be found form this state register.

Field	Name	Default	Access	Description
31: 2		0x0	RO	Reserved.
1	RXD_WORK	0x0	RO	The UART RXD is receiving.
0	TXD_WORK	0x0	RO	The UART TXD is transmitting.

I2C_MODE

- Name: I²C mode control register
- Address offset: 0x020

Description: The I²C can work in only one mode of two (master or slave) mode.

Field	Name	Default	Access	Description
31: 1		0x0	RO	Reserved.
0	MS_MODE	0x1	R/W	0x0: I ² CS mode. 0x1: I ² CM mode.

I2C_ADDR

- Name: I2C address register.
- Address offset: 0x024

Description: Both I²CM and I²CS use this register as I²CS address.

Field	Name	Default	Access	Description
31: 10		0x0	RO	Reserved.
9: 0	I2C_ADDR	0x133	R/W	The 10-bit I ² C slave address.

I2CM_CTRL

- Name: I²C master mode control register
- Address offset: 0x028

Description: The I²CM will implement I²C protocol in I²CM mode by configure this control register.

Field	Name	Default	Access	Description
31: 5		0	RO	Reserved.
4	GCALL	0x0	R/W	0x0: the first address is not general call address 0x1: the first address is general call address If user want to use the general call address, he has to clear the control bit of ADDR_MODE.
3	SBYTE	0x0	R/W	0x0: do not generate START byte. 0x1: generate START byte.
2	HS_MODE	0x0	R/W	High speed mode.
1	STOP	0x0	R/W	0: if TX FIFO is empty, the I ² CM holds SCL low. 1: if TX FIFO is empty, the I ² CM terminates transfer, then generating STOP condition.
0	ADDR_MODE	0x0	R/W	0: 7-bit address mode. 1: 10-bit address mode. If both of GCALL and ADDR_MODE is set by “1”, I ² CM only work in 10-bit I ² C address mode.

I2CM_CODE

- Name: I²C master code register.
- Address offset: 0x02C

Description: The I²C master code is used in high-speed mode.

Field	Name	Default	Access	Description
31: 3		0x0	RO	Reserved.
2: 0	MCODE	0x1	R/W	I ² CM code.

I2CS_CTRL

- Name: I²C slave control register
- Address offset: 0x030

Description: Many special I²C protocol can be configured by this control register.

Field	Name	Default	Access	Description
31: 1		0x0	RO	Reserved.
0	GCALL_MODE	0x0	R/W	0x0: receive general call data. 0x1: don't receive the data of general call.

I2C_FM_DIV

- Name: I²C fast-mode clock divider register.
- Address offset: 0x034

Description: The I²C fast mode divisor. this divisor is used in I²C master code and START byte transfer.

Field	Name	Default	Access	Description
31: 8		0x0	RO	Reserved.
7: 0	I2C_FM_DIV	0x5	R/W	The I ² C fast mode divisor.

I2C_HOLD

- Name: the hold time register.
- Address offset: 0x038

Description: This register controls the amount of hold time on the SDA signal after a negative of SCL in both master and slave mode.

Field	Name	Default	Access	Description
31: 8		0x0	RO	Reserved.
7: 0	I2C_HOLD	0x5	R/W	The I ² C hold time register. The hold time after SCL fall edge, used in both I ² CM and I ² CS. $T_{I2C_HOLD} = (I2C_HOLD + 1) * T_{PCLK}$ TPCLK: the PCLK period.

I2C_STA

- Name: I²C status register
- Address offset: 0x03C

Description: I²CM and I²CS state of working can be found in this state register.

Field	Name	Default	Access	Description
31: 9		0x0	RO	Reserved.
8	I2CS_WORK	0x0	RO	1'b0: I ² C slave is idle. 1'b1: I ² C slave is active.
7: 2		0x0	RO	Reserved.
1	I2CM_DATA	0x0	RO	If this bit equal 1'b1 that indicates I2C master is transmitting or receiving data.
0	I2CM_WORK	0x0	RO	1'b0: I ² C master is idle. 1'b1: I ² C master is active.

SPI_MODE

- Name: SPI (master or slave) mode select register.
- Address offset: 0x040



Description: The SPI master or slave can be selected by write this register.

Field	Name	Default	Access	Description
31: 1		0x0	RO	Reserved.
0	MS_MODE	0x1	R/W	0x0: SPI slave mode. 0x1: SPI master mode.

SPI_CTRL

- Name: SPI control register
- Address offset: 0x044

Description: User can control phase, polarity and data length by this control register.

Field	Name	Default	Access	Description
31: 10		0x0	RO	Reserved.
9	NSS_CTRL	0x0	R/W	Infect the NSS output 0x0: the output NSS is controlled by the NSS_TOGGLE 0x1: the output NSS is controlled by the NSS_DATA
8	NSS_TOGGL E	0x0	R/W	0x0: the NSS toggle at end of each data item. 0x1: the NSS is not change until the SPI master stop.
7	CPOL	0x0	R/W	Clock polarity 0x0: Inactive state of serial clock is low. 0x1: Inactive state of serial clock is high.
6	CPHA	0x0	R/W	Clock phase 0x0: Serial clock toggles in middle of first data bit. 0x1: Serial clock toggles at start of first data bit.
5: 4	TMOD	0x0	R/W	0x0: Transmit & Receive. 0x1: Transmit Only. 0x2: Receive Only. 0x3: TMOD keep the last set value unchanged.
3: 0	DATA_SIZE	0x7	R/W	Data frame size.

- DATA_SIZE

DATA_SIZE value	Description
0x0	Reserved. If user set this value, it will be replaced by the default value 0x7.
0x1	Reserved. If user set this value, it will be replaced by the default value 0x7.
0x2	Reserved. If user set this value, it will be replaced by the default value 0x7.



0x3	4-bit data length for each transfer.
0x4	5-bit data length for each transfer.
0x5	6-bit data length for each transfer.
0x6	7-bit data length for each transfer.
0x7	8-bit data length for each transfer.
0x8	9-bit data length for each transfer.
0x9	10-bit data length for each transfer.
0xA	11-bit data length for each transfer.
0xB	12-bit data length for each transfer.
0xC	13-bit data length for each transfer.
0xD	14-bit data length for each transfer.
0xE	15-bit data length for each transfer.
0xF	16-bit data length for each transfer.

SPI_STA

- Name: SPI status register
- Address offset: 0x048

Description: The SPI working status can be found in this register.

Field	Name	Default	Access	Description
31: 1		0x0	RO	Reserved.
0	SPI_WORKING	0x0	RO	SPI master or slave is working.

INTR_CTRL

- Name: interrupts control register.
- Address offset: 0x04C

Description: This register controls the threshold value for the transmit and receive FIFO memory.

Field	Name	Default	Access	Description
31: 17		0x0	RO	Reserved.
16	TH_MODE	0x0	R/W	0x1: not less than threshold. 0x0: not larger than threshold.
15: 11		0x0	RO	Reserved.

Field	Name	Default	Access	Description
10: 8	RX_FIFO_TH	0x1	R/W	RX FIFO threshold level control bits. If the received data in RX FIFO equal the threshold value, an interrupt will be generated. 0x0: threshold value keep the last set value unchanged. 0x1: 1-byte data 0x2: 2-byte data 0x3: 3-byte data 0x4: 4-byte data 0x5: 5-byte data 0x6: 6-byte data 0x7: 7-byte data
7: 3		0x0	RO	Reserved.
2: 0	TX_FIFO_TH	0x1	R/W	TX FIFO threshold control bits. 0x0: threshold value keep the last set value unchanged. 0x1: 1-byte 0x2: 2-byte 0x3: 3-byte 0x4: 4-byte 0x5: 5-byte 0x6: 6-byte 0x7: 7-byte

INTR_EN

- Name: interrupts enable control register.
- Address offset: 0x050

Description: the interrupt can be enable by set the control bit.

Field	Name	Default	Access	Description
31: 19		0x0	RO	Reserved.
18	SPI_STOP_EN	0x0	R/W	The interrupt enable control bit. 0x0: disable the interrupt. 0x1: enable the interrupt.
17	I2C_AERR_EN	0x0	R/W	
16	I2CS_GCALL_EN	0x0	R/W	
15	I2CM_LOSE_ARBI_EN	0x0	R/W	
14	I2C_NACK_EN	0x0	R/W	
13	I2C_STOP_EN	0x0	R/W	
12	UART_PERR_EN	0x0	R/W	
11	UART_RX_STOP_EN	0x0	R/W	
10	UART_TX_STOP_EN	0x0	R/W	

9	RX_WERR_EN	0x0	R/W
8	RX_RERR_EN	0x0	R/W
7	RX_FULL_EN	0x0	R/W
6	RX_EMPTY_EN	0x0	R/W
5	RX_THOLD_EN	0x0	R/W
4	TX_WERR_EN	0x0	R/W
3	TX_RERR_EN	0x0	R/W
2	TX_FULL_EN	0x0	R/W
1	TX_EMPTY_EN	0x0	R/W
0	TX_THOLD_EN	0x0	R/W

INTR_STA

- Name: interrupts status register.
- Address offset: 0x054

Description: which interrupt will be found in this register, all the state of the interrupt can be masked.

Field	Name	Default	Access	Description
31: 19		0x0	RO	Reserved.
18	SPI_STOP	0x0	RO	See “RAW_INTR_STA” on page 35 for a detailed description of these bits.
17	I2C_AERR	0x0	RO	
16	I2CS_GCALL	0x0	RO	
15	I2CM_LOSE_ARBI	0x0	RO	
14	I2C_NACK	0x0	RO	
13	I2C_STOP	0x0	RO	
12	UART_PERR	0x0	RO	
11	UART_RX_STOP	0x0	RO	
10	UART_TX_STOP	0x0	RO	
9	RX_WERR	0x0	RO	
8	RX_RERR	0x0	RO	
7	RX_FULL	0x0	RO	
6	RX_EMPTY	0x0	RO	
5	RX_THOLD	0x0	RO	
4	TX_WERR	0x0	RO	
3	TX_RERR	0x0	RO	
2	TX_FULL	0x0	RO	
1	TX_EMPTY	0x0	RO	
0	TX_THOLD	0x0	RO	

RAW_INTR_STA

- Name: interrupts status register.
- Address offset: 0x058

Description: The raw of interrupts status register.

Field	Name	Default	Access	Description
31: 19		0x0	RO	Reserved.
18	RAW_SPI_STOP	0x0	RO	SPI master: if the TX FIFO is empty, or the RX FIFO is full the interrupt will be generated before the FSM enter the SPIM_IDLE state. SPI slave: the interrupt will be generated after one data transfer.
17	RAW_I2C_AERR	0x0	RO	If the I2C_ADDR is loaded a reserved address or a special address in 7-bit address mode, the I2CM will generate a interrupt.
16	RAW_I2CS_GCALL	0x0	RO	When the I2CS received a general call, the interrupt will be generated.
15	RAW_I2CM_LOSE_A RBI	0x0	RO	When the I2CM loses arbitration, the interrupt will be generated.
14	RAW_I2C_NACK	0x0	RO	The interrupt will generated either the I2CM or I2CS receives NACK.
13	RAW_I2C_STOP	0x0	RO	The interrupt will be generated when the I2CM or I2CS finished working before the FSM enter the IDLE state.
12	RAW_UART_PERR	0x0	RO	The interrupt will be generated if the parity check fails when the UART RXD received the data.
11	RAW_UART_RX_STO P	0x0	RO	UART RXD: the interrupt will be generated after the data is received.
10	RAW_UART_TX_STO P	0x0	RO	UART TXD: the interrupt will be generated at the STOP bit.
9	RAW_RX_WERR	0x0	RO	The interrupt will be generated, when the RX FIFO is written after the RX FIFO is full.
8	RAW_RX_RERR	0x0	RO	The interrupt will be generated, when the RX FIFO is read after the RX FIFO is empty.
7	RAW_RX_FULL	0x0	RO	The interrupt will be generated when RX FIFO is full.
6	RAW_RX_EMPTY	0x0	RO	The interrupt will be generated when RX FIFO is empty.

Field	Name	Default	Access	Description
5	RAW_RX_THOLD	0x0	RO	TH_MODE = 0 (TH_MODE is field of INTR_CTRL register) The interrupt will be generated, when the RX FIFO depth of data not larger than threshold value of RX_FIFO_TH (the field in INTR_CTRL register). TH_MODE = 1 The interrupt will be generated, when the RX FIFO depth of data not less than threshold value of RX_FIFO_TH.
4	RAW_TX_WERR	0x0	RO	The interrupt will be generated, when the TX FIFO is written after the TX FIFO is full.
3	RAW_TX_RERR	0x0	RO	The interrupt will be generated, when the TX FIFO is read after the TX FIFO is empty.
2	RAW_TX_FULL	0x0	RO	The interrupt will be generated when TX FIFO is full.
1	RAW_TX_EMPTY	0x0	RO	The interrupt will be generated when TX FIFO is empty.
0	RAW_TX_THOLD	0x0	RO	TH_MODE = 0 The interrupt will be generated, when the TX FIFO depth of data not larger than threshold value of TX_FIFO_TH (the field in INTR_CTRL register). TH_MODE = 1 The interrupt will be generated, when the TX FIFO depth of data not less than threshold value of TX_FIFO_TH.

INTR_UNMASK

- Name: interrupts unmask register.
- Address offset: 0x05C

Description: the interrupt can be masked by this control register.

Field	Name	Default	Access	Description
31: 19		0x0	RO	Reserved.
18	SPI_STOP_MASK	0x0	R/W	Interrupt unmask register 0x0: masked
17	I2C_AERR_MASK	0x0	R/W	

Field	Name	Default	Access	Description
16	I2CS_GCALL_MASK	0x0	R/W	0x1: unmasked.
15	I2CM_LOSE_ARBI_MASK	0x0	R/W	
14	I2C_NACK_MASK	0x0	R/W	
13	I2C_STOP_MASK	0x0	R/W	
12	UART_PERR_MASK	0x0	R/W	
11	UART_RX_STOP_MASK	0x0	R/W	
10	UART_TX_STOP_MASK	0x0	R/W	
9	RX_WERR_MASK	0x0	R/W	
8	RX_RERR_MASK	0x0	R/W	
7	RX_FULL_MASK	0x0	R/W	
6	RX_EMPTY_MASK	0x0	R/W	
5	RX_THOLD_MASK	0x0	R/W	
4	TX_WERR_MASK	0x0	R/W	
3	TX_RERR_MASK	0x0	R/W	
2	TX_FULL_MASK	0x0	R/W	
1	TX_EMPTY_MASK	0x0	R/W	
0	TX_THOLD_MASK	0x0	R/W	

INTR_CLR

- Name: interrupts clear register.
- Address offset: 0x060

Description: the control register be use to clear interrupt.

Field	Name	Default	Access	Description
31: 19		0x0	RO	Reserved.
18	SPI_STOP_CLR	0x0	WO	Write 1'b1 to clear the RAW_SPI_STOP interrupt of the RAW_INTR_STA register.
17	I2C_AERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_I2C_AERR interrupt of the RAW_INTR_STA register.
16	I2CS_GCALL_CLR	0x0	WO	Write 1'b1 to clear the RAW_I2CS_GCALL interrupt of the RAW_INTR_STA register.
15	I2CM_LOSE_ARBI_CLR	0x0	WO	Write 1'b1 to clear the RAW_I2CM_LOSE_ARBI interrupt of the RAW_INTR_STA register.

Field	Name	Default	Access	Description
14	I2C_NACK_CLR	0x0	WO	Write 1'b1 to clear the RAW_I2C_NACK interrupt of the RAW_INTR_STA register.
13	I2C_STOP_CLR	0x0	WO	Write 1'b1 to clear the RAW_I2C_STOP interrupt of the RAW_INTR_STA register.
12	UART_PERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_UART_PERR interrupt of the RAW_INTR_STA register.
11	UART_RX_STOP_CLR	0x0	WO	Write 1'b1 to clear the RAW_UART_RX_STOP interrupt of the RAW_INTR_STA register.
10	UART_TX_STOP_CLR	0x0	WO	Write 1'b1 to clear the RAW_UART_TX_STOP interrupt of the RAW_INTR_STA register.
9	RX_WERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_RX_WERR interrupt of the RAW_INTR_STA register.
8	RX_RERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_RX_RERR interrupt of the RAW_INTR_STA register.
7	RX_FULL_CLR	0x0	WO	Write 1'b1 to clear the RAW_RX_FULL interrupt of the RAW_INTR_STA register.
6	RX_EMPTY_CLR	0x0	WO	Write 1'b1 to clear the RAW_RX_EMPTY interrupt of the RAW_INTR_STA register.
5	RX_THOLD_CLR	0x0	WO	Write 1'b1 to clear the RAW_RX_THOLD interrupt of the RAW_INTR_STA register.
4	TX_WERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_TX_WERR interrupt of the RAW_INTR_STA register.
3	TX_RERR_CLR	0x0	WO	Write 1'b1 to clear the RAW_TX_RERR interrupt of the RAW_INTR_STA register.
2	TX_FULL_CLR	0x0	WO	Write 1'b1 to clear the RAW_TX_FULL interrupt of the RAW_INTR_STA register.
1	TX_EMPTY_CLR	0x0	WO	Write 1'b1 to clear the RAW_TX_EMPTY interrupt of the RAW_INTR_STA register.

Field	Name	Default	Access	Description
0	TX_THOLD_CLR	0x0	WO	Write 1'b1 to clear the RAW_TX_THOLD interrupt of the RAW_INTR_STA register.

DMA_CTRL

- Name: DMA interface control register.
- Address offset: 0x064

Description: this register is design for controlling the TX FIFO or RX FIFO DMA interface.

Field	Name	Default	Access	Description
31: 2		0x0	RO	Reserved.
1	RX_DMA_EN	0x0	R/W	RX FIFO DMA interface enable. 0x0: disable the RX FIFO DMA interface. 0x1: enable the RX FIFO DMA interface.
0	TX_DMA_EN	0x0	R/W	TX FIFO DMA interface enable. 0x0: disable the TX FIFO DMA interface. 0x1: enable the TX FIFO DMA interface.

DMA_THRESHOLD

- Name: DMA interface threshold control register.
- Address offset: 0x068

Description: the threshold value for DMA trigger.

Field	Name	Default	Access	Description
31: 12		0x0	RO	Reserved.
11: 8	RX_DMA_TH	0x4	R/W	The counter is the threshold level for RX FIFO DMA interface. If the RX FIFO DMA interface is enable, and the number of received data words in RX FIFO not less than the threshold value, then the RX FIFO DMA interface requests that DMA read data from RX FIFO. 0x0: the threshold keep the last set value unchanged. 0x1: 1-data frame ... 0x5: 5-data frame 0x6: 6-data frame 0x7: 7-data frame 0x8: 8-data frame 0x9-0xF: Reserved.

7: 4		0x0	RO	Reserved.
3: 0	TX_DMA_TH	0x4	R/W	The counter is the threshold level for TX FIFO DMA interface. If the TX FIFO DMA interface is enable, and the number of stored data words in TX FIFO not larger than the threshold value, then the TX FIFO DMA interface requests that DMA write data to TX FIFO. 0x0: the threshold keep the last set value unchanged. 0x1: 1-data frame ... 0x5: 5-data frame 0x6: 6-data frame 0x7: 7-data frame 0x8: 8-data frame 0x9-0xF: Reserved.

SPI_NSS_DATA

- Name: SPI NSS data register.
- Address offset: 0x06C

Description: User can input NSS data by this nss data register.

Field	Name	Default	Access	Description
31: 1		0x0	RO	Reserved.
0	NSS_DATA	0x0	R/W	0x0: if NSS_CTRL is set 1, the “0” will be propagated to NSS 0x1: if NSS_CTRL is set 1, the “1” will be propagated to NSS.

5 Watchdog (WDT)

5.1 Register Description

5.1.1 Register Memory map

Table 5-1 Memory Map of WDT

Name	Address Offset	Width	Access	Reset value	Description
------	----------------	-------	--------	-------------	-------------



WDT_CR	0x00	5	R/W	5'h2	WDT control register
WDT_time_out	0x04	8	R/W	8'h0	WDT timeout range register
WDT_current_value	0x08	32	R	32'hffff	WDT current counter value register
WDT_restart	0x0c	8	W	8'h0	WDT counter restart register
WDT_int_status	0x10	1	R	1'b0	WDT interrupt status register
WDT_int_clr	0x14	1	R	1'b0	WDT interrupt clear register

5.1.2 Register Field Description

Control Register (WDT_CR)

Address Offset: 0x00

Read/write access: read/write

Table 5-2 WDT_CR Field Description

Bits	Name	R/W	Description
31:5	N/A		Reserved and read as zero (0).

4:2	RPL	R/W	This is used to select the number of pclk cycles for which the system reset stays asserted. The range of values available is 2 to 256 pclk cycles. 000 – 2 pclk cycles 001 – 4 pclk cycles 010 – 8 pclk cycles 011 – 16 pclk cycles 100 – 32 pclk cycles 101 – 64 pclk cycles 110 – 128 pclk cycles 111 – 256 pclk cycles Reset Value: 3'b0.
1	RMOD	R/W	0 = Generate a system reset. 1 = First generate an interrupt and if it is not cleared by the time a second timeout occurs then generate a system reset. Reset Value: 1
0	WDT_EN	R/W	Once this bit has been enabled, it can only be cleared by a system reset. 0 = WDT disabled. 1 = WDT enabled. Reset Value: 0

Timeout Range Register (WDT_time_out)

Address Offset: 0x04

Read/write access: read/write

Table 5-3 WDT_time_out Field Description

Bits	Name	R/W	Description
31:8	Reserved and read as zero (0).		

7:4	TOP_INIT	R/W	Used to select the timeout period that the watchdog counter restarts from for the first counter restart(kick). This register should be written after reset and before the WDT is enabled. A change of the TOP_INIT is seen only once the WDT has been enabled, and any change after the first kick is not seen as subsequent kicks use the period specified by the TOP bits. The range of values available for a 32-bit watchdog counter are: Where $i=TOP_INIT$ and $t=timeout\ period$ for $i = 0\ to\ 15$, $t=32'hffff$ Reset Value: 4'h0
3:0	TOP	R/W	Timeout period. This field is used to select the timeout period from which the watchdog counter restarts. A change of the timeout period takes effect only after the next counter restart (kick). The range of values available are: where $i = TOP$ and $t = timeout\ period$ for $i = 0\ to\ 15$, $t=32'hffff$ Reset Value: 4'h0

Current Counter Value Register (WDT_current_value)

Address Offset: 0x08

Read/write access: read

Table 5-4 WDT_current_value Field Description

Bits	Name	R/W	Description
31:0	Current Counter Value Register	R	This register, when read, is the current value of the internal counter. This value is read coherently whenever it is read. Reset Value: 32'hffff

Counter Restart Register (WDT_CRR)

Address Offset: 0x0c

Read/write access: write

Table 5-5 WDT_CRR Field Description

Bits	Name	R/W	Description
31:8	Reserved and read as zero.		
7:0	Counter Restart Register	W	This register is used to restart the WDT counter. As a safety feature to prevent accidental restarts, the value 0x76 must be written. A restart also clears the WDT interrupt. Reading this register returns zero. Reset Value: 0

Interrupt Status Register (WDT_int_status)

Address Offset: 0x10

Read/write access: read

Table 5-6 WDT_STAT Field Description

Bits	Name	R/W	Description
31:1	Reserved and read as zero.		
0	Interrupt Status Register	R	This register shows the interrupt status of the WDT. 1 = Interrupt is active regardless of polarity. 0 = Interrupt is inactive. Reset Value: 0

Interrupt Clear Register (WDT_int_clr)

Address Offset: 0x14

Read/write access: read

Table 5-7 WDT_int_clr Field Description

Bits	Name	R/W	Description
31:1	Reserved and read as zero.		
0	Interrupt Clear Register	R	Clears the watchdog interrupt. This can be used to clear the interrupt without restarting the watchdog counter. Reset Value: 0

5.2 Work Flow

This section describes the configuration of the WDR for a simple use. The special configuration should refer to the WDT register description.

Configure the WDT for use:

1. Disable the WDT by writing watchdog control register WDT_CR.
2. Configure the WDT_time_out.
3. Enable the WDT by writing the WDT_CR.

6 Pulse Width Modulation (PWM)

6.1 Register Description

6.1.1 Register Memory Map

Table 6-3 all internal registers

Name	Address Offset	Width	R/W	Description
PWMCFG	0x00	32 bits	R/W	PWM configure Reset Value: 0x0
PWMINVERTTRIG	0x04	32 bits	R/W	PWM signal is inverted Reset Value: 0x0
PWM01TRIG PWM23TRIG PWM45TRIG	0x08 0x0c 0x10	32 bits	R/W	Contain the trigger generate compare value Reset Value: 0x0
PWMINTEN1	0x14	32 bits	R/W	Interrupt enable for group2/1/0 Reset Value: 0x0
PWMINTEN2	0x18	32 bits	R/W	Interrupt enable for group5/4/3 Reset Value: 0x0
PWMRIS1	0x1c	32 bits	RO	Raw interrupt status for group2/1/0 Reset Value: 0x0
PWMRIS2	0x20	32 bits	RO	Raw interrupt status for group5/4/3 Reset Value: 0x0
PWMIC1	0x24	32 bits	R/W	Interrupt clear for group2/1/0 Reset Value: 0x0

PWMIC2	0x28	32 bits	R/W	Interrupt clear for group5/4/3 Reset Value: 0x0
PWMIS1	0x2c	32 bits	RO	Interrupt status for group2/1/0 Reset Value: 0x0
PWMIS2	0x30	32 bits	RO	Interrupt status for group5/4/3 Reset Value: 0x0
PWMCTL	0x34	32 bits	R/W	Configure the PWM generation blocks Reset Value: 0x0
PWM01LOAD PWM23LOAD PWM45LOAD	0x38 0x3c 0x40	32 bits	R/W	Contain the load value of the PWM counter Reset Value: 0x0
PWM01COUNT PWM23COUNT PWM45COUNT	0x44 0x48 0x4c	32 bits	RO	Contain the current value of the PWM counter Reset Value: 0x0
PWM0CMP PWM1CMP PWM2CMP PWM3CMP PWM4CMP PWM5CMP	0x50 0x54 0x58 0x5c 0x60 0x64	32 bits	R/W	Contain a value to be compared against the counter Reset Value: 0x0
PWM01DB PWM23DB PWM45DB	0x68 0x6c 0x70	32 bits	R/W	Contain the number of clock ticks to delay Reset Value: 0x0
CAPCTL	0x74	32 bits	R/W	Input capture control Reset Value: 0x0
CAPINTEN	0x78	32 bits	R/W	Input capture interrupt enable Reset Value: 0x0



CAPRIS	0x7c	32 bits	RO	Input capture raw interrupt status Reset Value: 0x0
CAPIC	0x80	32 bits	R/W	Input capture interrupt clear Reset Value: 0x0
CAPIS	0x84	32 bits	RO	Input capture interrupt status Reset Value: 0x0
CAP01T	0x88	32 bits	RO	Input capture counter value Reset Value: 0x0
CAP23T	0x8c			
CAP45T	0x90			
Cap01match	0x94	32 bits	R/W	Capture match value low 16bits Reset Value:0x0
Cap23match	0x98			
Cap45match	0x9c			
Tim_int_en	0xa0	32 bits	R/W	Tim int enable Reset Value:0x0
timris	0xa4	32 bits	RO	Timer interrupt status Reset Value:0x0
Tim_int_clr	0xa8	32 bits	R/W	Timer interrupt clear Reset Value:0x0
timis	0xac	32 bits	RO	Timer interrupt status Reset Value:0x0
Tim01load	0xb0	32 bits	R/W	Timer load value low 16bits Reset Value:0x0
Tim23load	0xb4			
Tim45load	0xb8			
Tim01count	0xbc	32 bits	RO	Timer count value low 16bits Reset Value:0x0
Tim23count	0xc0			
Tim45count	0xc4			



Cnt01val	0xc8			Current count value low 16bits Reset Value:0x0
Cnt23val	0xcc	32 bits	RO	
Cnv45val	0xd0			

6.1.2 Register Field Description

PWMCFG

This register configures the global operation of the PWM module.

- Name: PWM configuration
- Address Offset: 0x00

Table 6–1 PWMCFG Field Description

Bits	Name	R/W	Description
31:28	Reserved and read as zero.		
27	cntdiven	R/W	Counter frequency division configuration 1 : frequency division enable 0 : frequency division disable, so the counter clock source is system clock
26:24	cntdiv	R/W	Counter frequency division selection 111 : divide by 128 110 : divide by 128 101 : divide by 64 100 : divide by 32 011 : divide by 16 010 : divide by 8 001 : divide by 4 000 : divide by 2
23	tim5en	R/W	Group 5 output compare configuration 1 : output compare enable 0 : output compare disable
22	tim4en	R/W	Group 4 output compare configuration 1 : output compare enable 0 : output compare disable

21	tim3en	R/W	Group 3 output compare configuration 1 : output compare enable 0 : output compare disable
20	tim2en	R/W	Group 2 output compare configuration 1 : output compare enable 0 : output compare disable
19	tim1en	R/W	Group 1 output compare configuration 1 : output compare enable 0 : output compare disable
18	tim0en	R/W	Group 0 output compare configuration 1 : output compare enable 0 : output compare disable
17	cap5en	R/W	Group 5-channel 10 input capture configuration 1 : input capture enable 0 : input capture disable
16	cap4en	R/W	Group 4-channel 8 input capture configuration 1 : input capture enable 0 : input capture disable
15	cap3en	R/W	Group 3-channel 6 input capture configuration 1 : input capture enable 0 : input capture disable
14	cap2en	R/W	Group 2-channel 4 input capture configuration 1 : input capture enable 0 : input capture disable
13	cap1en	R/W	Group 1-channel 2 input capture configuration 1 : input capture enable 0 : input capture disable
12	cap0en	R/W	Group 0-channel 0 input capture configuration 1 : input capture enable 0 : input capture disable

11	pwm11en	R/W	Channel 11 PWM configuration 1 : PWM output enable 0 : PWM output disable
10	pwm10en	R/W	Channel 10 PWM configuration 1 : PWM output enable 0 : PWM output disable
9	pwm9en	R/W	Channel 9 PWM configuration 1 : PWM output enable 0 : PWM output disable
8	pwm8en	R/W	Channel 8 PWM configuration 1 : PWM output enable 0 : PWM output disable
7	pwm7en	R/W	Channel 7 PWM configuration 1 : PWM output enable 0 : PWM output disable
6	pwm6en	R/W	Channel 6 PWM configuration 1 : PWM output enable 0 : PWM output disable
5	pwm5en	R/W	Channel 5 PWM configuration 1 : PWM output enable 0 : PWM output disable
4	pwm4en	R/W	Channel 4 PWM configuration 1 : PWM output enable 0 : PWM output disable
3	pwm3en	R/W	Channel 3 PWM configuration 1 : PWM output enable 0 : PWM output disable
2	pwm2en	R/W	Channel 2 PWM configuration 1 : PWM output enable 0 : PWM output disable

1	pwm1en	R/W	Channel 1 PWM configuration 1 : PWM output enable 0 : PWM output disable
0	pwm0en	R/W	Channel 0 PWM configuration 1 : PWM output enable 0 : PWM output disable

PWMINVERTTRIG

This register controls the ADC trigger generation capabilities of the PWM generator (group) and provides a master control of the polarity of the PWM signals on the device pins.

- Name: PWM Output Inversion and trigger signal
- Address Offset: 0x04

Table 6–2 PWMINVERTTRIG Field Description

Bits	Name	R/W	Description
31:12	Reserved and read as zero.		
11	pwm11inv	R/W	Invert PWM11 signal (channel11) 1 : PWM5 signal is inverted 0 : PWM5 signal is not inverted
10	pwm10inv	R/W	Invert PWM10 signal (channel10) 1 : PWM4 signal is inverted 0 : PWM4 signal is not inverted
9	pwm9inv	R/W	Invert PWM9 signal (channel9) 1 : PWM3 signal is inverted 0 : PWM3 signal is not inverted
8	pwm8inv	R/W	Invert PWM8 signal (channel8) 1 : PWM2 signal is inverted 0 : PWM2 signal is not inverted
7	pwm7inv	R/W	Invert PWM7 signal (channel7) 1 : PWM1 signal is inverted 0 : PWM1 signal is not inverted

6	pwm6inv	R/W	Invert PWM6 signal (channel6) 1 : PWM0 signal is inverted 0 : PWM0 signal is not inverted
5	pwm5inv	R/W	Invert PWM5 signal (channel5) 1 : PWM5 signal is inverted 0 : PWM5 signal is not inverted
4	pwm4inv	R/W	Invert PWM4 signal (channel4) 1 : PWM4 signal is inverted 0 : PWM4 signal is not inverted
3	pwm3inv	R/W	Invert PWM3 signal (channel3) 1 : PWM3 signal is inverted 0 : PWM3 signal is not inverted
2	pwm2inv	R/W	Invert PWM2 signal (channel2) 1 : PWM2 signal is inverted 0 : PWM2 signal is not inverted
1	pwm1inv	R/W	Invert PWM1 signal (channel1) 1 : PWM1 signal is inverted 0 : PWM1 signal is not inverted
0	pwm0inv	R/W	Invert PWM0 signal (channel0) 1 : PWM0 signal is inverted 0 : PWM0 signal is not inverted

PWM01TRIG

This register contains the value generating trigger signals.

- Name: PWM group 0 and group 1 trigger compare value
- Address Offset: 0x08

Table 6–3 PWM01TRIG Field Description

Bits	Name	R/W	Description
31:16	pwm1trig	R/W	The value generating group 1 trigger signal
15:0	pwm0trig	R/W	The value generating group 0 trigger signal

PWM23TRIG



This register contains the value generating trigger signals.

- Name: PWM group 2 and group 3 trigger compare value
- Address Offset: 0x0c

Table 6–4 PWM23TRIG Field Description

Bits	Name	R/W	Description
31:16	pwm3trig	R/W	The value generating group 3 trigger signal
15:0	pwm2trig	R/W	The value generating group 2 trigger signal

PWM45TRIG

This register contains the value generating trigger signals.

- Name: PWM group 4 and group 5 trigger compare value
- Address Offset: 0x10

Table 6–5 PWM45TRIG Field Description

Bits	Name	R/W	Description
31:16	pwm5trig	R/W	The value generating group 5 trigger signal
15:0	pwm4trig	R/W	The value generating group 4 trigger signal

PWMINTEN1

This register controls the global interrupt generation capabilities of the PWM module.

- Name: PWM Interrupt enable for group 2/1/0
- Address Offset: 0x14

Table 6–6 PWMINTEN1 Field Description

Bits	Name	R/W	Description
31:30	Reserved and read as zero.		
29	Int2encmpbd	R/W	Interrupt for counter=comparator B Down (channel5) 1 : a raw interrupt occurs when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt

28	Int2encmpad	R/W	Interrupt for counter=comparator A Down (channel4) 1 : a raw interrupt occurs when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt
27	Int2encmpbu	R/W	Interrupt for counter=comparator B Up (channel5) 1 : a raw interrupt occurs when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
26	Int2encmpau	R/W	Interrupt for counter=comparator A Up (channel4) 1 : a raw interrupt occurs when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
25	Int2encntload	R/W	Interrupt for counter=Load (channel5/4) 1 : a raw interrupt occurs when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
24	Int2encntzero	R/W	Interrupt for counter=0 (channel5/4) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt
23:22	Reserved and read as zero.		
21	Int1encmpbd	R/W	Interrupt for counter=comparator B Down (channel3) 1 : a raw interrupt occurs when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt
20	Int1encmpad	R/W	Interrupt for counter=comparator A Down (channel2) 1 : a raw interrupt occurs when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt

19	Int1encmpbu	R/W	Interrupt for counter=comparator B Up (channel3) 1 : a raw interrupt occurs when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
18	Int1encmpau	R/W	Interrupt for counter=comparator A Up (channel2) 1 : a raw interrupt occurs when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
17	Int1encntload	R/W	Interrupt for counter=Load (channel3/2) 1 : a raw interrupt occurs when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
16	Int1encntzero	R/W	Interrupt for counter=0 (channel3/2) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		
13	Int0encmpbd	R/W	Interrupt for counter=comparator B Down (channel1) 1 : a raw interrupt occurs when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt
12	Int0encmpad	R/W	Interrupt for counter=comparator A Down (channel0) 1 : a raw interrupt occurs when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt
11	Int0encmpbu	R/W	Interrupt for counter=comparator B Up (channel1) 1 : a raw interrupt occurs when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt

10	Int0encmpau	R/W	Interrupt for counter=comparator A Up (channel0) 1 : a raw interrupt occurs when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt
9	Int0encntload	R/W	Interrupt for counter=Load (channel1/0) 1 : a raw interrupt occurs when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
8	Int0encntzero	R/W	Interrupt for counter=0 (channel1/0) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt
7:0	Reserved and read as zero.		

PWMINTEN2

This register controls the global interrupt generation capabilities of the PWM module.

- Name: PWM Interrupt enable for group 5/4/3
- Address Offset: 0x18

Table 6-7 PWMINTEN2 Field Description

Bits	Name	R/W	Description
31:22	Reserved and read as zero.		
21	Int5encmpbd	R/W	Interrupt for counter=comparator B Down (group5-channel11) 1 : a raw interrupt occurs when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt
20	Int5encmpad	R/W	Interrupt for counter=comparator A Down (group5-channel10) 1 : a raw interrupt occurs when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt

19	Int5encmpbu	R/W	Interrupt for counter=comparator B Up (group5-channel11) 1 : a raw interrupt occurs when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
18	Int5encmpau	R/W	Interrupt for counter=comparator A Up (group5-channel10) 1 : a raw interrupt occurs when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
17	Int5encntload	R/W	Interrupt for counter=Load (group5-channel11/10) 1 : a raw interrupt occurs when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
16	Int5encntzero	R/W	Interrupt for counter=0 (group5-channel11/10) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		
13	Int4encmpbd	R/W	Interrupt for counter=comparator B Down (group4-channel9) 1 : a raw interrupt occurs when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt
12	Int4encmpad	R/W	Interrupt for counter=comparator A Down (group4-channel8) 1 : a raw interrupt occurs when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt

11	Int4encmpbu	R/W	Interrupt for counter=comparator B Up (group4-channel9) 1 : a raw interrupt occurs when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
10	Int4encmpau	R/W	Interrupt for counter=comparator A Up (group4-channel8) 1 : a raw interrupt occurs when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
9	Int4encntload	R/W	Interrupt for counter=Load (group4-channel8/9) 1 : a raw interrupt occurs when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
8	Int4encntzero	R/W	Interrupt for counter=0 (group4-channel8/9) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt
7:6	Reserved and read as zero.		
5	Int3encmpbd	R/W	Interrupt for counter=comparator B Down (group3-channel7) 1 : a raw interrupt occurs when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt
4	Int3encmpad	R/W	Interrupt for counter=comparator A Down (group3-channel6) 1 : a raw interrupt occurs when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt

3	Int3encmpbu	R/W	Interrupt for counter=comparator B Up (group3-channel7) 1 : a raw interrupt occurs when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
2	Int3encmpau	R/W	Interrupt for counter=comparator A Up (group3-channel6) 1 : a raw interrupt occurs when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
1	Int3encntload	R/W	Interrupt for counter=Load (group3-channel7/6) 1 : a raw interrupt occurs when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
0	Int3encntzero	R/W	Interrupt for counter=0 (group3-channel7/6) 1 : a raw interrupt occurs when the counter is zero 0 : no interrupt

PWMRIS1

This register provides the current set of interrupt sources that asserted, regardless of whether they cause an interrupt to be asserted to the controller.

- Name: PWM raw interrupt status for group 2/1/0
- Address Offset: 0x1c

Table 6–8 PWMRIS1 Field Description

Bits	Name	R/W	Description
31:30	Reserved and read as zero.		
29	Intris2cmpbd	RO	Raw Interrupt for counter=comparator B Down (group2-channel5) 1 : a raw interrupt asserted when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt

28	Intris2cmpad	RO	Raw Interrupt for counter=comparator A Down (group2-channel4) 1 : a raw interrupt asserted when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt
27	Intris2cmpbu	RO	Raw Interrupt for counter=comparator B Up (group2-channel5) 1 : a raw interrupt asserted when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
26	Intris2cmpau	RO	Raw Interrupt for counter=comparator A Up (group2-channel4) 1 : a raw interrupt asserted when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
25	Intris2cntload	RO	Raw Interrupt for counter=Load (group2-channel5/4) 1 : a raw interrupt asserted when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
24	Intris2cntzero	RO	Raw Interrupt for counter=0 (group2-channel5/4) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt
23:22	Reserved and read as zero.		
21	Intris1cmpbd	RO	Raw Interrupt for counter=comparator B Down (group1-channel3) 1 : a raw interrupt asserted when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt

20	Intris1cmpad	RO	Raw Interrupt for counter=comparator A Down (group1-channel2) 1 : a raw interrupt asserted when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt
19	Intris1cmpbu	RO	Raw Interrupt for counter=comparator B Up (group1-channel3) 1 : a raw interrupt asserted when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
18	Intris1cmpau	RO	Raw Interrupt for counter=comparator A Up (group1-channel2) 1 : a raw interrupt asserted when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
17	Intris1cntload	RO	Raw Interrupt for counter=Load (group1-channel3/2) 1 : a raw interrupt asserted when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
16	Intris1cntzero	RO	Raw Interrupt for counter=0 (group1-channel3/2) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		
13	Intris0cmpbd	RO	Raw Interrupt for counter=comparator B Down (group0-channel1) 1 : a raw interrupt asserted when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt

12	Intris0cmpad	RO	Raw Interrupt for counter=comparator A Down (group0-channel0) 1 : a raw interrupt asserted when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt
11	Intris0cmpbu	RO	Raw Interrupt for counter=comparator B Up (group0-channel1) 1 : a raw interrupt asserted when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt
10	Intris0cmpau	RO	Raw Interrupt for counter=comparator A Up (group0-channel0) 1 : a raw interrupt asserted when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt
9	Intris0cntload	RO	Raw Interrupt for counter=Load (group0-channel1/0) 1 : a raw interrupt asserted when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
8	Intris0cntzero	RO	Raw Interrupt for counter=0 (group0-channel1/0) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt
7:0	Reserved and read as zero.		

PWMRIS2

This register controls the global interrupt generation capabilities of the PWM module.

- Name: PWM raw interrupt status for group 5/4/3
- Address Offset: 0x20

Table 6–9 PWMRIS2 Field Description

Bits	Name	R/W	Description
31:22	Reserved and read as zero.		

21	Intris5cmpbd	RO	Raw Interrupt for counter=comparator B Down (group5-channel11) 1 : a raw interrupt asserted when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt
20	Intris5cmpad	RO	Raw Interrupt for counter=comparator A Down (group5-channel10) 1 : a raw interrupt asserted when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt
19	Intris5cmpbu	RO	Raw Interrupt for counter=comparator B Up (group5-channel11) 1 : a raw interrupt asserted when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
18	Intris5cmpau	RO	Raw Interrupt for counter=comparator A Up (group5-channel10) 1 : a raw interrupt asserted when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
17	Intris5cntload	RO	Raw Interrupt for counter=Load (group5-channel11/10) 1 : a raw interrupt asserted when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
16	Intris5cntzero	RO	Raw Interrupt for counter=0 (group5-channel11/10) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		

13	Intris4cmpbd	RO	Raw Interrupt for counter=comparator B Down (group4-channel9) 1 : a raw interrupt asserted when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt
12	Intris4cmpad	RO	Raw Interrupt for counter=comparator A Down (group4-channel8) 1 : a raw interrupt asserted when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt
11	Intris4cmpbu	RO	Raw Interrupt for counter=comparator B Up (group4-channel9) 1 : a raw interrupt asserted when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
10	Intris4cmpau	RO	Raw Interrupt for counter=comparator A Up (group4-channel8) 1 : a raw interrupt asserted when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
9	Intris4cntload	RO	Raw Interrupt for counter=Load (group4-channel9/8) 1 : a raw interrupt asserted when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
8	Intris4cntzero	RO	Raw Interrupt for counter=0 (group4-channel9/8) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt
7:6	Reserved and read as zero.		

5	Intris3cmpbd	RO	Raw Interrupt for counter=comparator B Down (group3-channel7) 1 : a raw interrupt asserted when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt
4	Intris3cmpad	RO	Raw Interrupt for counter=comparator A Down (group3-channel6) 1 : a raw interrupt asserted when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt
3	Intris3cmpbu	RO	Raw Interrupt for counter=comparator B Up (group3-channel7) 1 : a raw interrupt asserted when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
2	Intris3cmpau	RO	Raw Interrupt for counter=comparator A Up (group3-channel6) 1 : a raw interrupt asserted when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
1	Intris3cntload	RO	Raw Interrupt for counter=Load (group3-channel7/6) 1 : a raw interrupt asserted when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
0	Intris3cntzero	RO	Raw Interrupt for counter=0 (group3-channel7/6) 1 : a raw interrupt asserted when the counter is zero 0 : no interrupt

PWMIC1

This register clears the interrupt.

- Name: PWM interrupt clear for group 2/1/0
- Address Offset: 0x24

Table 6–10 PWMIC1 Field Description

Bits	Name	R/W	Description
31:30	Reserved and read as zero.		
29	Intic2cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group2-channel5) 1 : clear 0 : not clear
28	Intic2cmpad	R/W	Clear Interrupt for counter=comparator A Down (group2-channel4) 1 : clear 0 : not clear
27	Intic2cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group2-channel5) 1 : clear 0 : not clear
26	Intic2cmpau	R/W	Clear Interrupt for counter=comparator A Up (group2-channel4) 1 : clear 0 : not clear
25	Intic2cntload	R/W	Clear Interrupt for counter=Load (group2-channel5/4) 1 : clear 0 : not clear
24	Intic2cntzero	R/W	Clear Interrupt for counter=0 (group2-channel5/4) 1 : clear 0 : not clear
23:22	Reserved and read as zero.		
21	Intic1cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group1-channel3) 1 : clear 0 : not clear

20	Intic1cmpad	R/W	Clear Interrupt for counter=comparator A Down (group1-channel2) 1 : clear 0 : not clear
19	Intic1cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group1-channel3) 1 : clear 0 : not clear
18	Intic1cmpau	R/W	Clear Interrupt for counter=comparator A Up (group1-channel2) 1 : clear 0 : not clear
17	Intic1cntload	R/W	Clear Interrupt for counter=Load (group1-channel3/2) 1 : clear 0 : not clear
16	Intic1cntzero	R/W	Clear Interrupt for counter=0 (group1-channel3/2) 1 : clear 0 : not clear
15:14	Reserved and read as zero.		
13	Intic0cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group0-channel1/0) 1 : clear 0 : not clear
12	Intic0cmpad	R/W	Clear Interrupt for counter=comparator A Down (group0-channel0) 1 : clear 0 : not clear
11	Intic0cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group0-channel1) 1 : clear 0 : not clear

10	Intic0cmpau	R/W	Clear Interrupt for counter=comparator A Up (group0-channel0) 1 : clear 0 : not clear
9	Intic0cntload	R/W	Clear Interrupt for counter=Load (group0-channel1/0) 1 : clear 0 : not clear
8	Intic0cntzero	R/W	Clear Interrupt for counter=0 (group0-channel1/0) 1 : clear 0 : not clear
7:0	Reserved and read as zero.		

PWMIC2

This register clears the interrupt.

- Name: PWM interrupt clear for group 5/4/3
- Address Offset: 0x28

Table 6–11 PWMIC2 Field Description

Bits	Name	R/W	Description
31:22	Reserved and read as zero.		
21	Intic5cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group5-channel11) 1 : clear 0 : not clear
20	Intic5cmpad	R/W	Clear Interrupt for counter=comparator A Down (group5-channel10) 1 : clear 0 : not clear
19	Intic5cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group5-channel11) 1 : clear 0 : not clear

18	Intic5cmpau	R/W	Clear Interrupt for counter=comparator A Up (group5-channel10) 1 : clear 0 : not clear
17	Intic5cntload	R/W	Clear Interrupt for counter=Load (group5-channel11/10) 1 : clear 0 : not clear
16	Intic5cntzero	R/W	Clear Interrupt for counter=0 (group5-channel11/10) 1 : clear 0 : not clear
15:14	Reserved and read as zero.		
13	Intic4cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group4-channel9) 1 : clear 0 : not clear
12	Intic4cmpad	R/W	Clear Interrupt for counter=comparator A Down (group4-channel8) 1 : clear 0 : not clear
11	Intic4cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group4-channel9) 1 : clear 0 : not clear
10	Intic4cmpau	R/W	Clear Interrupt for counter=comparator A Up (group4-channel8) 1 : clear 0 : not clear
9	Intic4cntload	R/W	Clear Interrupt for counter=Load (group4-channel9/8) 1 : clear 0 : not clear

8	Intic4cntzero	R/W	Clear Interrupt for counter=0 (group4-channel9/8) 1 : clear 0 : not clear
7:6	Reserved and read as zero.		
5	Intic3cmpbd	R/W	Clear Interrupt for counter=comparator B Down (group3-channel7) 1 : clear 0 : not clear
4	Intic3cmpad	R/W	Clear Interrupt for counter=comparator A Down (group3-channel6) 1 : clear 0 : not clear
3	Intic3cmpbu	R/W	Clear Interrupt for counter=comparator B Up (group3-channel7) 1 : clear 0 : not clear
2	Intic3cmpau	R/W	Clear Interrupt for counter=comparator A Up (group3-channel6) 1 : clear 0 : not clear
1	Intic3cntload	R/W	Clear Interrupt for counter=Load (group3-channel7/6) 1 : clear 0 : not clear
0	Intic3cntzero	R/W	Clear Interrupt for counter=0 (group3-channel7/6) 1 : clear 0 : not clear

PWMIS1

This register provides the current set of interrupt sources that asserted, regardless of whether they cause an interrupt to be asserted to the controller.

- Name: PWM interrupt status for group 2/1/0
- Address Offset: 0x2c

Table 6–12 PWMIS1 Field Description

Bits	Name	R/W	Description
31:30	Reserved and read as zero.		
29	Intis2cmpbd	RO	Interrupt for counter=comparator B Down (group2-channel5) 1 : a interrupt asserted when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt
28	Intis2cmpad	RO	Interrupt for counter=comparator A Down (group2-channel4) 1 : a interrupt asserted when the counter matches the value in the PWM2CMP register value when counting down 0 : no interrupt
27	Intis2cmpbu	RO	Interrupt for counter=comparator B Up (group2-channel5) 1 : a interrupt asserted when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
26	Intis2cmpau	RO	Interrupt for counter=comparator A Up (group2-channel4) 1 : a interrupt asserted when the counter matches the value in the PWM2CMP register value when counting up 0 : no interrupt
25	Intis2cntload	RO	Interrupt for counter=Load (group2-channel5/4) 1 : a interrupt asserted when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
24	Intis2cntzero	RO	Interrupt for counter=0 (group2-channel5/4) 1 : a interrupt asserted when the counter is zero 0 : no interrupt
23:22	Reserved and read as zero.		

21	Intis1cmpbd	RO	Interrupt for counter=comparator B Down (group1-channel3) 1 : a interrupt asserted when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt
20	Intis1cmpad	RO	Interrupt for counter=comparator A Down (group1-channel2) 1 : a interrupt asserted when the counter matches the value in the PWM1CMP register value when counting down 0 : no interrupt
19	Intis1cmpbu	RO	Interrupt for counter=comparator B Up (group1-channel3) 1 : a interrupt asserted when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
18	Intis1cmpau	RO	Interrupt for counter=comparator A Up (group1-channel2) 1 : a interrupt asserted when the counter matches the value in the PWM1CMP register value when counting up 0 : no interrupt
17	Intis1cntload	RO	Interrupt for counter=Load (group1-channel3/2) 1 : a interrupt asserted when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
16	Intis1cntzero	RO	Interrupt for counter=0 (group1-channel3/2) 1 : a interrupt asserted when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		
13	Intis0cmpbd	RO	Interrupt for counter=comparator B Down (group0-channel1) 1 : a interrupt asserted when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt

12	Intis0cmpad	RO	Interrupt for counter=comparator A Down (group0-channel0) 1 : a interrupt asserted when the counter matches the value in the PWM0CMP register value when counting down 0 : no interrupt
11	Intis0cmpbu	RO	Interrupt for counter=comparator B Up (group0-channel1) 1 : a interrupt asserted when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt
10	Intis0cmpau	RO	Interrupt for counter=comparator A Up (group0-channel0) 1 : a interrupt asserted when the counter matches the value in the PWM0CMP register value when counting up 0 : no interrupt
9	Intis0cntload	RO	Interrupt for counter=Load (group0-channel1/0) 1 : a interrupt asserted when the counter matches the values in the PWM01LOAD register value 0 : no interrupt
8	Intis0cntzero	RO	Interrupt for counter=0 (group0-channel1/0) 1 : a interrupt asserted when the counter is zero 0 : no interrupt
7:0	Reserved and read as zero.		

PWMIS2

This register controls the global interrupt generation capabilities of the PWM module.

- Name: PWM raw interrupt status for group 5/4/3
- Address Offset: 0x30

Table 6–13 PWMIS2 Field Description

Bits	Name	R/W	Description
31:22	Reserved and read as zero.		

21	Intis5cmpbd	RO	Interrupt for counter=comparator B Down (group5-channel11) 1 : a interrupt asserted when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt
20	Intis5cmpad	RO	Interrupt for counter=comparator A Down (group5-channel10) 1 : a interrupt asserted when the counter matches the value in the PWM5CMP register value when counting down 0 : no interrupt
19	Intis5cmpbu	RO	Interrupt for counter=comparator B Up (group5-channel11) 1 : a interrupt asserted when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
18	Intis5cmpau	RO	Interrupt for counter=comparator A Up (group5-channel10) 1 : a interrupt asserted when the counter matches the value in the PWM5CMP register value when counting up 0 : no interrupt
17	Intis5cntload	RO	Interrupt for counter=Load (group5-channel11/10) 1 : a interrupt asserted when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
16	Intis5cntzero	RO	Interrupt for counter=0 (group5-channel11/10) 1 : a interrupt asserted when the counter is zero 0 : no interrupt
15:14	Reserved and read as zero.		
13	Intis4cmpbd	RO	Interrupt for counter=comparator B Down (group4-channel9) 1 : a interrupt asserted when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt

12	Intis4cmpad	RO	Interrupt for counter=comparator A Down (group4-channel8) 1 : a interrupt asserted when the counter matches the value in the PWM4CMP register value when counting down 0 : no interrupt
11	Intis4cmpbu	RO	Interrupt for counter=comparator B Up (group4-channel9) 1 : a interrupt asserted when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
10	Intis4cmpau	RO	Interrupt for counter=comparator A Up (group4-channel8) 1 : a interrupt asserted when the counter matches the value in the PWM4CMP register value when counting up 0 : no interrupt
9	Intis4cntload	RO	Interrupt for counter=Load (group4-channel9/8) 1 : a interrupt asserted when the counter matches the values in the PWM45LOAD register value 0 : no interrupt
8	Intis4cntzero	RO	Interrupt for counter=0 (group4-channel9/8) 1 : a interrupt asserted when the counter is zero 0 : no interrupt
7:6	Reserved and read as zero.		
5	Intis3cmpbd	RO	Interrupt for counter=comparator B Down (group3-channel7) 1 : a interrupt asserted when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt
4	Intis3cmpad	RO	Interrupt for counter=comparator A Down (group3-channel6) 1 : a interrupt asserted when the counter matches the value in the PWM3CMP register value when counting down 0 : no interrupt

3	Intis3cmpbu	RO	Interrupt for counter=comparator B Up (group3-channel7) 1 : a interrupt asserted when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
2	Intis3cmpau	RO	Interrupt for counter=comparator A Up (group3-channel6) 1 : a interrupt asserted when the counter matches the value in the PWM3CMP register value when counting up 0 : no interrupt
1	Intis3cntload	RO	Interrupt for counter=Load (group3-channel7/6) 1 : a interrupt asserted when the counter matches the values in the PWM23LOAD register value 0 : no interrupt
0	Intis3cntzero	RO	Interrupt for counter=0 (group3-channel7/6) 1 : a interrupt asserted when the counter is zero 0 : no interrupt

PWMCTL

This register configures the PWM counter mode.

- Name: PWM control
- Address Offset: 0x34

Table 6-14 PWMCTL Field Description

Bits	Name	R/W	Description
31:18	Reserved and read as zero.		
17:16	Sync5mode	R/W	Synchronize mode (group5-channel11/10) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update

15:14	Sync4mode	R/W	Synchronize mode (group4-channel9/8) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update
13:12	Sync3mode	R/W	Synchronize mode (group3-channel7/6) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update
11:10	sync2mode	R/W	Synchronize mode (group2-channel5/4) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update
9:8	sync1mode	R/W	Synchronize mode (group1-channel3/2) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update
7:6	sync0mode	R/W	Synchronize mode (group0-channel1/0) 00 : update register value when counter reach to zero 01 : update register value when counter reach to load value 10 : update register value when counter reach to zero and load value 11 : not update

5	pwm5mode	R/W	Counter mode (group5-channel11/10) 1 : count-up/down mode 0 : count-up mode
4	pwm4mode	R/W	Counter mode (group4-channel9/8) 1 : count-up/down mode 0 : count-up mode
3	pwm3mode	R/W	Counter mode (group3-channel7/6) 1 : count-up/down mode 0 : count-up mode
2	pwm2mode	R/W	Counter mode (group2-channel5/4) 1 : count-up/down mode 0 : count-up mode
1	pwm1mode	R/W	Counter mode (group1-channel3/2) 1 : count-up/down mode 0 : count-up mode
0	pwm0mode	R/W	Counter mode (group0-channel1/0) 1 : count-up/down mode 0 : count-up mode

PWM01LOAD

PWM23LOAD

PWM45LOAD

This register contains the load value for the PWM counter.

- Name: PWM load value
- Address Offset: 0x38, 0x3c, 0x40

Table 6-15 PWMnmLOAD Field Description

Bits	Name	R/W	Description
31:16	loadm	R/W	Counter load m value (m=1 : group1-channel3/2 ; m=3 : group3-channel7/6 ; m=5 : group5-channel11/10) The counter load value

15:0	loadn	R/W	Counter load n value (n=0 : group0-channel1/0 ; n=2 : group2-channel5/4 ; n=4 : group4-channel9/8) The counter load value
------	-------	-----	---

PWM01COUNT

PWM23COUNT

PWM45COUNT

This register contains the current value of the PWM counter.

- Name: PWM current value
- Address Offset: 0x44, 0x48, 0x4c

Table 6–16 PWMnmCOUNT Field Description

Bits	Name	R/W	Description
31:16	countm	RO	Counter m value (m=1 : group1-channel3/2 ; m=3 : group3-channel7/6 ; m=5 : group5-channel11/10) The current value of the counter
15:0	countn	RO	Counter n value (n=0 : group0-channel1/0 ; n=2 : group2-channel5/4 ; n=4 : group4-channel9/8) The current value of the counter

PWM0CMP

PWM1CMP

PWM2CMP

PWM3CMP

PWM4CMP

PWM5CMP

This register contain a value to be compared against the counter.

- Name: PWM compare value
- Address Offset: 0x50, 0x54, 0x58, 0x5c, 0x60, 0x64

Table 6–17 PWMnCMP Field Description

Bits	Name	R/W	Description
31:16	compnb	R/W	Comparator B value (n=0 : group0-channel1; n=1 : group1-channel3 ; n=2 : group2-channel5 ; n=3 : group2-channel7 ; n=4 : group4-channel9 ; n=5 : group5-channel11) The value to be compared against the counter
15:0	compna	R/W	Comparator A value (n=0 : group0-channel0 ; n=1 : group1-channel2 ; n=2 : group2-channel4 ; n=3 : group2-channel6 ; n=4 : group4-channel8 ; n=5 : group5-channel10) The value to be compared against the counter

PWM01DB
PWM23DB
PWM45DB

These registers contains the number of clock ticks to delay.

- Name: PWM deadband value
- Address Offset: 0x68, 0x6c, 0x70

Table 6–18 PWMnmDB Field Description

Bits	Name	R/W	Description
31:26	Reserved and read as zero		

25	dbmen	R/W	Dead-band m generator (group) enable (m=1 : group1-channel3/2 ; m=3 : group3-channel7/6 ; m=5 : group5-channel11/10) 1 : insert dead-band into the output signals 0 : pass the PWM signals through simply
24	dbnen	R/W	Dead-band n generator (group) enable (n=0 : group0-channel1/0 ; n=2 : group2-channel5/4 ; n=4 : group4-channel9/8) 1 : insert dead-band into the output signals 0 : pass the PWM signals through simply
23:12	delaym	R/W	Dead-band delay (m=1 : group1-channel3/2 ; m=3 : group3-channel7/6 ; m=5 : group5-channel11/10) The number of clock tick to delay
11:0	delayn	R/W	Dead-band delay (n=0 : group0-channel1/0 ; n=2 : group2-channel5/4 ; n=4 : group4-channel9/8) The number of clock tick to delay

CAPCTL

This register configures input capture mode.

- Name: Input capture mode control
- Address Offset: 0x74

Table 6–19 CAPCTL Field Description

Bits	Name	R/W	Description
31:18	Reserved and read as zero		

17:16	cap5event	R/W	Capture5 edge event mode (group5-channel10) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge
15:14	cap4event	R/W	Capture4 edge event mode (group4-channel8) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge
13:12	cap3event	R/W	Capture3 edge event mode (group3-channel6) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge
11:10	cap2event	R/W	Capture2 edge event mode (group2-channel4) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge
9:8	cap1event	R/W	Capture1 edge event mode (group1-channel2) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge
7:6	cap0event	R/W	Capture0 edge event mode (group0-channel0) 00: posedge edge 01 : negedge edge 10 : Reserved and read as zero 11 : both edge

5	cap5mode	R/W	Capture5 mode (group5-channel10) 1 : edge count mode 0 : edge time mode
4	cap4mode	R/W	Capture4 mode (group4-channel8) 1 : edge count mode 0 : edge time mode
3	cap3mode	R/W	Capture3 mode (group3-channel6) 1 : edge count mode 0 : edge time mode
2	cap2mode	R/W	Capture2 mode (group2-channel4) 1 : edge count mode 0 : edge time mode
1	cap1mode	R/W	Capture1 mode (group1-channel2) 1 : edge count mode 0 : edge time mode
0	cap0mode	R/W	Capture0 mode (group0-channel0) 1 : edge count mode 0 : edge time mode

CAPINTEN

This register enables the interrupt for input capture function mode.

- Name: interrupt enable for input capture mode
- Address Offset: 0x78

Table 6–20 CAPINTEN Field Description

Bits	Name	R/W	Description
31:12	Reserved and read as zero.		
11	cap5timie	R/W	Capture5 edge time interrupt enable (group5-channel10) 1 : interrupt enable 0 : interrupt disable

10	cap4timie	R/W	Capture4 edge time interrupt enable (group4-channel8) 1 : interrupt enable 0 : interrupt disable
9	cap3timie	R/W	Capture3 edge time interrupt enable (group3-channel6) 1 : interrupt enable 0 : interrupt disable
8	cap2timie	R/W	Capture2 edge time interrupt enable (group2-channel4) 1 : interrupt enable 0 : interrupt disable
7	cap1timie	R/W	Capture1 edge time interrupt enable (group1-channel2) 1 : interrupt enable 0 : interrupt disable
6	cap0timie	R/W	Capture0 edge time interrupt enable (group0-channel0) 1 : interrupt enable 0 : interrupt disable
5	cap5cntie	R/W	Capture5 edge count interrupt enable (group5-channel10) 1 : interrupt enable 0 : interrupt disable
4	cap4cntie	R/W	Capture4 edge count interrupt enable (group4-channel8) 1 : interrupt enable 0 : interrupt disable

3	cap3cntie	R/W	Capture3 edge count interrupt enable (group3-channel6) 1 : interrupt enable 0 : interrupt disable
2	cap2cntie	R/W	Capture2 edge count interrupt enable (group2-channel4) 1 : interrupt enable 0 : interrupt disable
1	cap1cntie	R/W	Capture1 edge count interrupt enable (group1-channel2) 1 : interrupt enable 0 : interrupt disable
0	cap0cntie	R/W	Capture0 edge count interrupt enable (group0-channel0) 1 : interrupt enable 0 : interrupt disable

CAPRIS

This register shows the state of raw interrupt for input capture function mode.

- Name: raw interrupt status for input capture mode
- Address Offset: 0x7c

Table 6–21 CAPRIS Field Description

Bits	Name	R/W	Description
31:12	Reserved and read as zero.		
11	cap5timris	RO	Capture5 edge time raw interrupt occurs (group5-channel10) 1 : interrupt 0 : no interrupt

10	cap4timris	RO	Capture4 edge time raw interrupt occurs (group4-channel8) 1 : interrupt 0 : no interrupt
9	cap3timris	RO	Capture3 edge time raw interrupt occurs (group3-channel6) 1 : interrupt 0 : no interrupt
8	cap2timris	RO	Capture2 edge time raw interrupt occurs (group2-channel4) 1 : interrupt 0 : no interrupt
7	cap1timris	RO	Capture1 edge time raw interrupt occurs (group1-channel2) 1 : interrupt 0 : no interrupt
6	cap0timris	RO	Capture0 edge time raw interrupt occurs (group0-channel0) 1 : interrupt 0 : no interrupt
5	cap5cntris	RO	Capture5 edge count raw interrupt occurs (group5-channel10) 1 : interrupt 0 : no interrupt
4	cap4cntris	RO	Capture4 edge count raw interrupt occurs (group4-channel8) 1 : interrupt 0 : no interrupt

3	cap3cntris	RO	Capture3 edge count raw interrupt occurs (group3-channel6) 1 : interrupt 0 : no interrupt
2	cap2cntris	RO	Capture2 edge count raw interrupt occurs (group2-channel4) 1 : interrupt 0 : no interrupt
1	cap1cntris	RO	Capture1 edge count raw interrupt occurs (group1-channel2) 1 : interrupt 0 : no interrupt
0	cap0cntris	RO	Capture0 edge count raw interrupt occurs (group0-channel0) 1 : interrupt 0 : no interrupt

CAPIC

This register clears internal interrupt.

- Name: interrupt clear for input capture mode
- Address Offset: 0x80

Table 6–22 CAPIC Field Description

Bits	Name	R/W	Description
31:12	Reserved and read as zero.		
11	cap5timic	R/W	Clear Capture5 edge time interrupt (group5-channel10) 1 : clear interrupt 0 : not clear interrupt

10	cap4timic	R/W	Clear Capture4 edge time interrupt (group4-channel8) 1 : clear interrupt 0 : not clear interrupt
9	cap3timic	R/W	Clear Capture3 edge time interrupt (group3-channel6) 1 : clear interrupt 0 : not clear interrupt
8	cap2timic	R/W	Clear Capture2 edge time interrupt (group2-channel4) 1 : clear interrupt 0 : not clear interrupt
7	cap1timic	R/W	Clear Capture1 edge time interrupt (group1-channel2) 1 : clear interrupt 0 : not clear interrupt
6	cap0timic	R/W	Clear Capture0 edge time interrupt (group0-channel0) 1 : clear interrupt 0 : not clear interrupt
5	cap5cntic	R/W	Clear Capture5 edge count interrupt (group5-channel10) 1 : clear interrupt 0 : not clear interrupt
4	cap4cntic	R/W	Clear Capture4 edge count interrupt (group4-channel8) 1 : clear interrupt 0 : not clear interrupt

3	cap3cntic	R/W	Clear Capture3 edge count interrupt (group3-channel6) 1 : clear interrupt 0 : not clear interrupt
2	cap2cntic	R/W	Clear Capture2 edge count interrupt (group2-channel4) 1 : clear interrupt 0 : not clear interrupt
1	cap1cntic	R/W	Clear Capture1 edge count interrupt (group1-channel2) 1 : clear interrupt 0 : not clear interrupt
0	cap0cntic	R/W	Clear Capture0 edge count interrupt (group0-channel0) 1 : clear interrupt 0 : not clear interrupt

CAPIS

This register shows the state of interrupt for input capture function mode.

- Name: interrupt status for input capture mode
- Address Offset: 0x84

Table 6-23 CAPIS Field Description

Bits	Name	R/W	Description
31:12	Reserved and read as zero.		
11	cap5timis	RO	Capture5 edge time interrupt occurs (group5-channel10) 1 : interrupt 0 : no interrupt

10	cap4timis	RO	Capture4 edge time interrupt occurs (group4-channel8) 1 : interrupt 0 : no interrupt
9	cap3timis	RO	Capture3 edge time interrupt occurs (group3-channel6) 1 : interrupt 0 : no interrupt
8	cap2timis	RO	Capture2 edge time interrupt occurs (group2-channel4) 1 : interrupt 0 : no interrupt
7	cap1timis	RO	Capture1 edge time interrupt occurs (group1-channel2) 1 : interrupt 0 : no interrupt
6	cap0timis	RO	Capture0 edge time interrupt occurs (group0-channel0) 1 : interrupt 0 : no interrupt
5	cap5cntis	RO	Capture5 edge count interrupt occurs (group5-channel10) 1 : interrupt 0 : no interrupt
4	cap4cntis	RO	Capture4 edge count interrupt occurs (group4-channel8) 1 : interrupt 0 : no interrupt

3	cap3cntis	RO	Capture3 edge count interrupt occurs (group3-channel6) 1 : interrupt 0 : no interrupt
2	cap2cntis	RO	Capture2 edge count interrupt occurs (group2-channel4) 1 : interrupt 0 : no interrupt
1	cap1cntis	RO	Capture1 edge count interrupt occurs (group1-channel2) 1 : interrupt 0 : no interrupt
0	cap0cntis	RO	Capture0 edge count interrupt occurs (group0-channel0) 1 : interrupt 0 : no interrupt

CAP01T
CAP23T
CAP45T

This register contains the counter value when edge match for input capture function mode.

- Name: captured current value for input capture function mode
- Address Offset: 0x88, 0x8c, 0x90

Table 6–24 CAPnmT Field Description

Bits	Name	R/W	Description
31:16	CAPnmTm	RO	Input capture m counter value (m=1 : group1-channel2 ; m=3 : group3-channel6 ; m=5 : group5-channel10)

15:0	CAPnmTn	RO	Input capture n counter value (n=0 : group1-channel0 ; n=2 : group3-channel4 ; n=4 : group5-channel8)
------	---------	----	--

CAP01MATCH

CAP23MATCH

CAP45MATCH

This register contains the match value for input capture function mode.

- Name: capture match value for input capture function mode
- Address Offset: 0x94, 0x98, 0x9c

Table 6–25 CAPnmMATCH Field Description

Bits	Name	R/W	Description
31:16	CAPnmMATCH _m	R/W	Input capture m match value (m=1 : group1-channel2 ; m=3 : group3-channel6 ; m=5 : group5-channel10)
15:0	CAPnmMATCH _n	R/W	Input capture n match value (n=0 : group1-channel0 ; n=2 : group3-channel4 ; n=4 : group5-channel8)

TIMINTEN

This register configures the interrupt enable for timer function mode.

- Name: interrupt enable for timer mode
- Address Offset: 0xa0

Table 6–26 TIMINTEN Field Description

Bits	Name	R/W	Description
31:6	Reserved and read as zero.		
5	tim5ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group5) 1 : interrupt enable 0 : interrupt disable

4	tim4ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group4) 1 : interrupt enable 0 : interrupt disable
3	tim3ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group3) 1 : interrupt enable 0 : interrupt disable
2	tim2ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group2) 1 : interrupt enable 0 : interrupt disable
1	tim1ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group1) 1 : interrupt enable 0 : interrupt disable
0	tim0ie	R/W	A interrupt enable when counter match the CMPnmMATCH value. (group0) 1 : interrupt enable 0 : interrupt disable

TIMRIS

This register shows the state of raw internal interrupt for timer function mode.

- Name: raw interrupt status for timer mode
- Address Offset: 0xa4

Table 6–27 TIMRIS Field Description

Bits	Name	R/W	Description
31:6	Reserved and read as zero.		
5	tim5ris	RO	Timer5 raw interrupt occurs. (group5) 1 : interrupt 0 : no interrupt

4	tim4ris	RO	Timer4 raw interrupt occurs. (group4) 1 : interrupt 0 : no interrupt
3	tim3ris	RO	Timer3 raw interrupt occurs. (group3) 1 : interrupt 0 : no interrupt
2	tim2ris	RO	Timer2 raw interrupt occurs. (group2) 1 : interrupt 0 : no interrupt
1	tim1ris	RO	Timer1 raw interrupt occurs. (group1) 1 : interrupt 0 : no interrupt
0	tim0ris	RO	Timer0 raw interrupt occurs. (group0) 1 : interrupt 0 : no interrupt

TIMIC

This register shows the interrupt clear for timer function mode.

- Name: interrupt clear for timer mode
- Address Offset: 0xa8

Table 6–28 TIMIC Field Description

Bits	Name	R/W	Description
31:6	Reserved and read as zero.		
5	tim5ic	R/W	Timer5 interrupt. (group5) 1 : clear interrupt 0 : not clear interrupt
4	tim4ic	R/W	Timer4 interrupt. (group4) 1 : clear interrupt 0 : not clear interrupt

3	tim3ic	R/W	Timer3 interrupt. (group3) 1 : clear interrupt 0 : not clear interrupt
2	tim2ic	R/W	Timer2 interrupt. (group2) 1 : clear interrupt 0 : not clear interrupt
1	tim1ic	R/W	Timer1 interrupt. (group1) 1 : clear interrupt 0 : not clear interrupt
0	tim0ic	R/W	Timer0 interrupt. (group0) 1 : clear interrupt 0 : not clear interrupt

TIMIS

This register shows the status of interrupt for timer function mode.

- Name: interrupt status for timer mode
- Address Offset: 0xac

Table 6–29 TIMIS Field Description

Bits	Name	R/W	Description
31:6	Reserved and read as zero.		
5	tim5is	RO	Timer5 interrupt occurs. (group5) 1 : interrupt 0 : no interrupt
4	tim4is	RO	Timer4 interrupt occurs. (group4) 1 : interrupt 0 : no interrupt
3	tim3is	RO	Timer3 interrupt occurs. (group3) 1 : interrupt 0 : no interrupt

2	tim2is	RO	Timer2 interrupt occurs. (group2) 1 : interrupt 0 : no interrupt
1	tim1is	RO	Timer1 interrupt occurs. (group1) 1 : interrupt 0 : no interrupt
0	tim0is	RO	Timer0 interrupt occurs. (group0) 1 : interrupt 0 : no interrupt

TIM01LOAD

TIM23LOAD

TIM45LOAD

This register contains the load value for timer function mode.

- Name: load value for timer function mode
- Address Offset: 0xb0, 0xb4, 0xb8

Table 6–30 TIMnmLOAD Field Description

Bits	Name	R/W	Description
31:16	timloadm	R/W	Timer m load value (m=1 : group1 ; m=3 : group3; m=5 : group5)
15:0	timloadn	R/W	Timer n load value (n=0 : group1 ; n=2 : group3 ; n=4 : group5)

TIM01COUNT

TIM23COUNT

TIM45COUNT

This register contains the current value for timer function mode.

- Name: current value for timer function mode
- Address Offset: 0xbc, 0xc0, 0xc4

Table 6–31 TIMnmCOUNT Field Description

Bits	Name	R/W	Description
31:16	timcntm	RO	Timer m current count value (m=1 : group1 ; m=3 : group3; m=5 : group5)
15:0	timcntn	RO	Timer n current count value (n=0 : group1 ; n=2 : group3 ; n=4 : group5)

CNT01VAL
CNT23VAL
CNT45VAL

This register contains the captured input pulse number value for input capture function mode.

- Name: captured value for input capture function mode
- Address Offset: 0xc8, 0xcc, 0xd0

Table 6–32 CNTnmVAL Field Description

Bits	Name	R/W	Description
31:16	Cntmval	RO	Counter m captured input pulse number value (m=1 : group1 ; m=3 : group3; m=5 : group5)
15:0	Cntnval	RO	Counter n captured input pulse number value (n=0 : group1 ; n=2 : group3 ; n=4 : group5)

6.2 Work Flow

The following example shows how to initialize the PWM generator 0 with a 25-KHz frequency, and with a 75% duty cycles on the PWM0 pin and a 25% duty on the

PWM1 pin. This example assumes the bus clock is 40-MHz.

1. Select the PWM count mode, for example, write 0x0 to PWMCTL (count up mode)
2. Set the PWM period. For a 25-KHz frequency, the period = 40us. The APB clock is 40-MHz and divided by 2, so that the PWM clock source is 20-MHz. This translates to 800 clock ticks per period. Use this value to set PWM0LOAD register. Write 0x320 to PWM0LOAD[31:16] and PWM0LOAD[15:0]. 20Mhz = 50ns
3. Set the pulse width of the PWM0 pin for a 75% duty cycle. Write 0xc8 to PWM0CMP[15:0].
4. Set the pulse width of the PWM1 pin for a 25% duty cycle. Write 0x258 to PWM0CMP[31:16].
5. Bypass Dead-Band insert, write 0x0 to PWM01DB.
6. Set the PWM clock frequency division and start the counter and enable output PWM signal, write 0x80000003 to PWMCFG.

7 Real-Time Clock (RTC)

7.1 Register Description

7.1.1 Register Memory Map

The following table describes the register memory map.

Table 7-1 RTC Memory Map

Name	Address offset	Width	Access	Reset Value	Description
RTC_current_value	0x00	32	R	0x0	Current Counter Value Register
RTC_match_value	0x04	32	RW	0x0	Counter Match Register
RTC_load_value	0x08	32	RW	0x0	Counter Load Register
RTC_CCR	0x0C	4~2bits	RW	0x0	Counter Control Register
RTC_int_status	0x10	32	R	0x0	Interrupt Status Register
RTC_raw_int_status	0x14	32	R	0x0	Interrupt Raw Status Register
RTC_int_clr	0x18	32	R	0x0	End of Interrupt Register

RTC_COMP_VERSION	0x1C	32	R	0x0	Component Version Register
RTC_DIV	0x20	20	RW	0x4000	Rtc clock divider value

7.1.2 Registers Field Description

Current Counter Value Register (RTC_current_value)

Address Offset: 0x00

Read/write access: read-only

Table 7-2 RTC_current_value Field Description

Bits	Name	R/W	Description
31:0	Current Counter Value	R	When read, this register is the current value of the internal counter. This value always is read coherently. Reset Value: 0x0

Counter Match Register (RTC_match_value)

Address Offset: 0x04

Read/write access: read/write

Table 7-3 RTC_match_value Field Description

Bits	Name	R/W	Description
31:0	Counter Match	R/W	Interrupt Match Register. When the internal counter matches this register, an interrupt is generated, provided interrupt generation is enabled. When appropriate, this value is written coherently. Only when all the bytes are written is the register used by the interrupt detection logic. Reset Value: 0x0

Counter Load Register (RTC_load_value)

Address Offset: 0x08

Read/write access: read/write

Table 7-4 RTC_CLR Field Description

Bits	Name	R/W	Description
------	------	-----	-------------

31:0	Counter Load	R/W	Loaded into the counter as the loaded value, which is written coherently. Reset Value: 0x0
------	--------------	-----	--

Counter Control Register (RTC_CCR)

Address Offset: 0x0C

Read/write access: read/write

Table 7-5 RTC_CCR Fiesld Description

Bits	Name	R/W	Description
31:4	N/A	N/A	Reserved and read as 0
3	rtc_wen	R/W	Allows the user to force the counter to wrap when a match occurs instead of waiting until the maximum count is reached. 0 = Wrap disabled 1 = Wrap enabled Reset Value: 0x0
2	Rtc_en	R/W	Allows the user to control counting in the counter. 0 = Counter disabled 1 = Counter enabled Reset Value: 0x0
1	rtc_mask	R/W	Allows the user to mask interrupt generation. 0 = Interrupt unmasked 1 = Interrupt masked Reset Value: 0x0
0	rtc_ien	R/W	Allows the user to disable interrupt generation. 0 = Interrupt disabled 1 = Interrupt enabled Reset Value: 0x0

Interrupt Status Register (RTC_int_status)

Address Offset: 0x10

Read/write access: read-only

Table 7-6 RTC_int_status Field Description

Bits	Name	R/W	Description
31:1	N/A	N/A	Reserved and read as 0
0	rtc_stat	R	This register is the masked raw status 0 = Interrupt is inactive 1 = Interrupt is active (regardless of polarity) Reset Value: 0x0

Interrupt Raw Status Register (RTC_Rawint_status)

Address Offset: 0x14

Read/write access: read-only

Table 7-7 RTC_Rawint_status Field Description

Bits	Name	R/W	Description
31:1	N/A	N/A	Reserved and read as 0
0	rtc_rstat	R	0 = Interrupt is inactive 1 = Interrupt is active (regardless of polarity) Reset Value: 0x0

End of Interrupt Register (RTC_int_clr)

Address Offset: 0x18

Read/write access: read-only

Table 7-8 RTC_int_clr Field Description

Bits	Name	R/W	Description
31:1	N/A	N/A	Reserved and read as 0
0	rtc_int_clr	R	By reading this location, the match interrupt is cleared. Performing read-to-clear on interrupts, the interrupt is cleared at the end of the read. Reset Value: 0x0

CLK DIV Register (RTC_DIV)

Address Offset: 0x20

Read/write access: read/write

Table 7-9 RTC_DIV Field Description

Bits	Name	R/W	Description
31:1	N/A	N/A	Reserved and read as 0
19:0	rtc_div	R/W	Rtc clcok divider base on this register value Reset Value: 0x4000

7.2 Work Flow

This section describe the configuration of the RTC for a simple use. The special configuration should refer to the RTC register description.

Configuration the RTC for use:

1. Set the control register (RTC_CCR) to reset value to disable the RTC first.
2. Set the RTC counter match register (RTC_match_value).
3. Set the RTC counter load register (RTC_load_value).
4. Set the RTC counter control register to the corresponding value (RTC_CCR).

8 General-purpose I/O (GPIO)

8.1 Register description

8.1.1 Register Memory Map

Table 8-1 Address Map of GPIO Registers

Name	Description	Offset	Initial Value	R/W	Access Size/bit
gpio_output_data	GPIO output data register	0x00	0x0	R/W	32
gpio_direction	GPIO direction register	0x04	0x0	R/W	32
gpio_ctl	GPIO control register	0x08	0x0	R/W	32
gpio_inten	Interrupt enable register	0x30	0x0	R/W	32
gpio_intmask	Interrupt mask register	0x34	0x0	R/W	32
gpio_inttype_level	Interrupt level register	0x38	0x0	R/W	32



gpio_int_polarity	Interrupt polarity register	0x3c	0x0	R/W	32
gpio_intstatus	Interrupt status of GPIO	0x40	0x0	R	32
gpio_rawintstatus	Raw interrupt status of GPIO (premasking)	0x44	0x0	R	32
gpio_porta_int_clr	GPIO clear interrupt register	0x4c	0x0	W	32
gpio_input_data	GPIO input data register	0x50	0x0	R	32

8.1.2 Registers Filed Description

Gpio_output_data:

Name: GPIO Data Register

Size: 32 bits

Address Offset: 0x00

Read/write access: read/write

Table 8-2 gpio_output_data Field Description

Bits	Field Name	Description	Initial value
31:0	GPIO Data Register	Values written to this register are output on the I/O signals. For GPIO if the corresponding data direction bits for GPIO are set to Output mode and the corresponding control bit for GPIO is set to Software mode. The value read back is equal to the last value written to this register.	32'b0

gpio_direction:

Name: GPIO Data Direction Register

Size: 32 bits

Address Offset: 0x04

Read/write access: read/write

Table 8-3 gpio_direction Field Description

Bits	Field Name	Description	Initial value
------	------------	-------------	---------------

31:0	GPIO Data Direction Register	Values written to this register independently control the direction of the corresponding data bit in GPIO. 0 – Input 1 – Output for each bit	32'b0
------	------------------------------	---	-------

gpio_ctl:

Name: GPIO data source register

Size: 32 bits

Address Offset: 0x08

Read/write access: read/write

Table 8-4 gpio_ctl Field Description

Bits	Field Name	Description	Initial value
31:0	GPIO Data Source Register	The data and control source for a signal can come from either software or hardware; this bit selects between them. 0-Software mode for each bit 1-Hardware mode for each bit Reset Value: 28'h0	32'b0

gpio_inten:

Name: Interrupt enable

Size: 32 bits

Address Offset: 0x30

Read/write access: read/write

Table 8-5 gpio_inten Field Description

Bits	Field Name	Description	Initial value
31:0	Interrupt enable	Allows each bit of GPIO to be configured for interrupts. By default the generation of interrupts is disabled. Whenever a 1 is written to a bit of this register, it configures the corresponding bit on GPIO to become an interrupt; otherwise, GPIO operates as a normal GPIO signal. Interrupts are disabled on the corresponding bits of GPIO if the corresponding data direction register is set to Output or if GPIO mode is set to Hardware. 0 – Configure GPIO bit as normal GPIO signal 1 – Configure GPIO bit as interrupt	32'b0

gpio_intmask:

Name: Interrupt mask

Size: 32 bits

Address Offset: 0x34

Read/write access: read/write

Table 8-6 gpio_intmask Field Description

Bits	Field Name	Description	Initial value
31:0	Interrupt mask	Controls whether an interrupt on GPIO can create an interrupt for the interrupt controller by not masking it. By default, all interrupts bits are unmasked. Whenever a 1 is written to a bit in this register, it masks the interrupt generation capability for this signal; otherwise interrupts are allowed through. The unmasked status can be read as well as the resultant status after masking. 0 – Interrupt bits are unmasked (default) 1 – Mask interrupt	32'b0

gpio_inttype_level

Name: Interrupt level

Size: 32 bits

Address Offset: 0x38

Read/write access: read/write

Table 8-7 gpio_inttype_level Field Description

Bits	Field Name	Description	Initial value
31:0	Interrupt level	Controls the type of interrupt that can occur on GPIO. Whenever a 0 is written to a bit of this register, it configures the interrupt type to be level-sensitive; otherwise, it is edge-sensitive. 0 – Level-sensitive (default) 1 – Edge-sensitive Reset Value:0x0	32'b0

gpio_int_polarity:

Name: Interrupt polarity



Size: 32 bits

Address Offset: 0x3c

Read/write access: read/write

Table 8-8 gpio_int_polarity Field Description

Bits	Field Name	Description	Initial value
31:0	Interrupt polarity	Controls the polarity of edge or level sensitivity that can occur on input of GPIO. Whenever a 0 is written to a bit of this register, it configures the interrupt type to falling-edge or active-low sensitive; otherwise, it is rising-edge or active-high sensitive. 0 – Active-low (default) 1 – Active-high	32'b0

gpio_intstatus:

Name: Interrupt status

Size: 32 bits

Address Offset: 0x40

Read/write access: read only

Table 8-9 gpio_intstatus Field Description

Bits	Field Name	Description	Initial value
31:0	Interrupt status	Interrupt status of GPIO	32'b0

gpio_rawintstatus:

Name: Raw interrupt status

Size: 32 bits

Address Offset: 0x44

Read/write access: read only

Table 8-10 gpio_rawintstatus Field Description

Bits	Field Name	Description	Initial value
31:0	Raw interrupt status	Raw interrupt of status of GPIO (pre-masking bits)	32'b0

gpio_int_clr:

Name: Clear interrupt

Size: 32 bits

Address Offset: 0x4c

Read/write access: write only

Table 8-11 gpio_int_clr Field Description

Bits	Field Name	Description	Initial value
31:0	Clear interrupt	Controls the clearing of edge type interrupts from GPIO. When a 1 is written into a corresponding bit of this register, the interrupt is cleared. All interrupts are cleared when GPIO is not configured for interrupts. 0 – No interrupt clear (default) 1 – Clear interrupt	32'b0

gpio_input_data:

Name: External GPIO

Size: 32 bits

Address Offset: 0x50

Read/write access: read only

Table 8-12 gpio_input_data Field Description

Bits	Field Name	Description	Initial value
31:0	External GPIO	When GPIO is configured as Input, then reading this location reads the values on the signal. When the data direction of GPIO is set as Output, reading this location reads the data register for GPIO.	32'b0

8.1.3 Work Flow

Configure GPIO output :

1. Set GPIO. gpio_swporta_ctl. gpio_swporta_ctl[0] = 1'b1, configure GPIO[0] use software mode
2. Set GPIO. gpio_swporta_ddr. gpio_swporta_ddr[0] = 1'b1, configure GPIO[0] direct is input
3. Set GPIO. gpio_swporta_dr. gpio_swporta_dr[0] = 1'b1, write data 1'b1 to gpio
4. Monitor PAD GPIO[0] == 1

Configure GPIO input:

1. Set GPIO. gpio_swporta_ctl. gpio_swporta_ctl[0] = 1'b1, configure GPIO[0] use software mode
2. Set GPIO. gpio_swporta_ddr. gpio_swporta_ddr[1] = 1'b0, configure GPIO[0] direct is output
3. Connect PAD GPIO[0] to high level
4. Read GPIO. GPIO external port register. GPIO external port register[0] == 1. Check GPIO[0] input data = 1

